INNOPOLIS UNIVERSITY

Bachelor of Science in Computer Science

ViKi: Learning Robotic Manipulation from Multiview Human Demonstrations

Bachelor's Graduation Thesis

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Abstract

Collecting robot demonstrations across different embodiments and observation viewpoints requires substantial teleoperation effort. Human manipulation recordings offer a reusable source of natural task motion, but converting them into robot observation–action pairs requires accurate perception, embodiment-specific retargeting, and consistent visual scene representation.

This thesis investigates ViKi, an offline pipeline combining multiview RGB-D human demonstration capture, morphology-aware motion retargeting, and 3D Gaussian Splatting (3DGS)-based scene rendering. The proposed workflow reconstructs hand and object motion from synchronized cameras, derives robot trajectories subject to kinematic and collision constraints, and renders candidate robot demonstrations from novel viewpoints with controlled appearance variations. The objective is to reuse one human demonstration across multiple robot embodiments, camera viewpoints, and scene variations

without collecting separate teleoperated data for each configuration.

The evaluation framework separates hand-tracking accuracy, object pose estimation error, cross-embodiment retargeting feasibility, novel-view rendering fidelity, and synthetic dataset diversity metrics. HO-Cap recordings with ground-truth MANO annotations provide external reference for perception assessment. This study focuses on dataset-generation pipeline design and validation; policy training, task success metrics, and real-robot deployment require subsequent work.

Keywords: robot learning, 3D Gaussian Splatting, human demonstrations, RGB-D perception, motion retargeting, multi-embodiment, dataset generation

Глава 1

Introduction

1.1 Background

Vision-based robot learning benefits from demonstrations that cover different objects, robot embodiments, camera viewpoints, and scene conditions. Direct teleoperation provides robot actions paired with observations, but acquiring additional recordings requires operator time and access to the relevant hardware. The cost depends on the collection interface and task; it need not grow in a fixed proportion to the number of robots. Demonstration reuse and synthetic data generation offer ways to reduce repeated collection [1], [2].

Human demonstrations provide another source of manipulation motion. A person can perform a task while cameras record hand–object interaction, without controlling a robot during capture. Such recordings do not directly supply robot joint states or executable actions. Their use requires estimating motion, identifying task-relevant interactions, and adapting the resulting trajectory to the target robot [3], [4].

3D Gaussian Splatting (3DGS) supports novel-view synthesis from calibrated images and provides an editable representation of scene appearance [5]. Prior systems already use this representation for robot-data augmentation and human-to-robot data generation [2], [6], [7]. This thesis investigates a ViKi workflow that combines markerless human-motion reconstruction from multiview RGB-D recordings with embodiment-specific retargeting and 3DGS-based rendering. Here, markerless refers to tracking the demonstrator without worn markers or instrumented gloves; calibration targets may still be used to establish camera geometry.

1.2 Problem Statement

The objective is to reuse a recorded human manipulation episode to generate candidate robot observation–action sequences under several embodiment and viewing configurations. Three connected requirements define the problem. Perception must recover hand and object motion in a consistent metric frame. Retargeting must adapt task-relevant motion to the robot's reach, joint limits, end-effector geometry, and the geometric constraints represented in the model. Rendering must produce observations consistent with the retargeted robot, moving objects, and selected camera pose.

The intended expansion is described by $\boldsymbol{M}$ robot embodiments, $\boldsymbol{K}$ viewpoints, and $\boldsymbol{P}$ appearance configurations. Their Cartesian product contains up to $\boldsymbol{MKP}$ candidate episode configurations per source demonstration. It is not a guarantee of $\boldsymbol{MKP}$ feasible or independent demonstrations. Some robots may not reach the task, some views may reveal unobserved surfaces, and some edits may invalidate visual or geometric assumptions. The study therefore needs to report both the requested combinations and the subset that passes explicit checks.

Stage-level measurements alone are insufficient. A low hand-reconstruction

error, an admissible inverse-kinematics solution, and a plausible rendered image can still disagree at their interfaces. The generated episode must also preserve frame conventions, timestamps, object motion, and the correspondence between rendered robot state and exported action. Geometric consistency does not, by itself, establish force closure, successful manipulation, or usefulness for policy learning.

1.3 Research Gap

The relevant literature includes integrated systems, not only isolated components. RoboSplat augments a robot demonstration across object poses, object types, camera views, scene appearance, lighting, and embodiments [6]. Real2Render2Real extracts object motion from a human video and generates robot data using reconstructed assets [2]. RoboPaint combines multiview RGB-D human capture, glove-based retargeting, and hybrid rendering with a 3DGS background [7]. SplatSim generates synthetic observations for simulated demonstrations, while GaussianFactory synthesises demonstrations from a scene scan using grasp and trajectory planning [8], [9].

Building on these systems, this thesis develops and assesses a particular

markerless multiview RGB-D route: how reliably can reconstructed hand–object motion be reused across target robots and rendered observation conditions, and where does this reuse fail? The proposed contribution lies in the implementation and controlled characterisation of that route, including the effects of capture geometry, perception error, robot morphology, and rendering choices. Chapter 2 compares the input requirements and representations of the closest systems.

1.4 Research Questions

The questions below define the intended evaluation of the pipeline. The implementation and results in Chapter 4 provide the evidence for answering them; questions without the required experiments remain open. A selected cohort of 28 HO-Cap manipulation sequences with eight RGB-D views provides the source material for the perception assessment; it is not the full dataset or its official held-out benchmark split.

RQ1: Hand Reconstruction Accuracy

How accurately does landmark detection, multiview reconstruction, and temporal regularisation recover human hand motion relative to the published

HO-Cap annotations?

Measures: MPJPE, P95 joint error, PCK@20mm, and prediction coverage, with directly observed and temporally filled predictions reported separately.

Reference: HO-Cap's semi-automatic MANO-based annotations [10]. These are reference labels, not a competing baseline or an independent motion-capture measurement. A parametric representation used by the reference does not imply that ViKi fits the same model.

RQ2: Object Tracking Accuracy

How accurately can rigid-object motion be recovered, and how does the result depend on segmentation and initial pose information?

Measures: Translation and rotation error, their aggregation level, and tracking coverage. Orientation scoring must account for declared object symmetries or report the limitations of the chosen pose convention.

Protocol: Separate autonomous tracking from tracking with supplied instance masks. Reference-assisted first-frame alignment measures subsequent relative drift, not absolute localisation. Multiview agreement is a self-consistency measure unless checked against an independent reference.

RQ3: Cross-Embodiment Retargeting Feasibility

For which target robots can the recovered trajectories satisfy stated kinematic and geometric constraints, and what tracking error remains?

Measures: Accepted-episode and accepted-frame rates, TCP translation and rotation error, joint and velocity-limit violations, self-collision margin, and clearance from explicitly modelled scene geometry. Solver convergence is reported separately from constraint satisfaction.

Candidate embodiments: UR10, UR5, Franka Panda, Kinova Gen3, and xArm7. The evaluation must state the actual robot models, base placements, end-effectors, and trajectories tested. Kinematic acceptance does not certify successful execution of the manipulation.

RQ4: Gripper State Transfer

How consistently can observed thumb–index motion and interaction timing be mapped to a parallel-jaw gripper command?

Measures: Aperture tracking or command error against an explicitly defined mapping, opening and closing timing relative to annotated events, actuation-limit violations, and infeasible grasp-width cases. A mapped command

is not a measurement of achieved contact.

Scope: Parallel-jaw transfer, initially using Robotiq 2F-85, is the primary case. Robotiq 3F and Allegro Hand require richer hand correspondences and a separate retargeting protocol; a scalar pinch signal does not determine their complete joint configurations.

RQ5: Object-Aware Retargeting

Does incorporating object motion improve geometric interaction consistency relative to hand-only inverse kinematics?

Comparison: Hand-only and object-relative retargeting on the same source episodes and robot configurations; a contact-aware variant can be included where a defined contact model is available.

Measures: Relative hand/gripper–object pose error, unintended penetration, and consistency of designated contact regions and timing. These are geometric interaction measures, not physical grasp-success rates. Replaying an imposed object trajectory cannot establish that the robot would cause that motion.

RQ6: 3DGS Reconstruction and Novel-View Fidelity

How do the number and placement of capture cameras affect reconstruction and novel-view quality for the scene components represented by 3DGS?

Measures: PSNR, SSIM, and valid-pixel depth RMSE on held-out views, with separate evaluation of background and task-relevant object regions.

Protocol: Fix train/test views and temporal splits before tuning. The rendered reference must depict the same scene state. A static background model must not be evaluated as if it reproduced the moving human and objects. Timing perturbations, if evaluated, are a separate controlled ablation.

RQ7: Robot Compositing Consistency

How consistently can a retargeted robot and moving objects be rendered within the reconstructed scene across time and camera viewpoints?

Measures: Depth-ordering and occlusion errors, boundary artifacts, temporal stability, and agreement between rendered state and exported robot state. Paired image metrics such as LPIPS require a corresponding reference of the same target scene; a human demonstration frame is not a ground-truth image of the replacement robot.

Comparison: Alternative compositing methods on identical trajectories, camera poses, and scene states. Human ratings, if collected, require a stated sampling and rating protocol. Visual plausibility alone does not certify correct interaction.

RQ8: Coverage and Yield of Dataset Expansion

What coverage of embodiment, viewpoint, and appearance configurations can be generated from each source demonstration while satisfying the defined kinematic, geometric, and data-consistency checks?

Measures: Requested and accepted episode counts, acceptance rate by factor combination, viewpoint coverage, failure categories, and generation time. Feature-space distances can describe visual variation under a fixed encoder, but are not evidence of additional task skills or learning benefit.

Comparison: Single-configuration export and controlled expansion of the same source episodes. A teleoperation cost comparison requires measured collection effort and an explicit accounting of calibration, scanning, annotation, computation, and rejected outputs.

The evaluation concerns reconstruction, retargeting, rendering, and their integration. Policy training, task-success trials in simulation or on hardware,

and real-robot deployment are outside this study. Claims about learning effectiveness or physically executable demonstrations require those additional experiments.

1.5 Intended Contributions

The planned contributions concern the implementation of the workflow and its experimental characterisation:

1. An offline pipeline design that separates multiview human-motion reconstruction, rigid-object tracking, robot retargeting, and visual synthesis through explicit intermediate representations.

2. A workflow for reusing a source demonstration across robot, viewpoint, and appearance configurations, with rejected cases and configuration-dependent assumptions recorded rather than hidden in the nominal expansion count.

3. An evaluation protocol for perception error, embodiment-specific feasibility, novel-view fidelity, and observation–action consistency, including controlled comparisons where reference data are available.

4. A characterisation of the resulting dataset coverage and generation effort within the tested tasks and embodiments.

The intended outcome is a dataset-generation workflow with measured operating limits, acceptance criteria, and documented failure cases.

1.6 Thesis Structure

Chapter 2 reviews human-motion representations, retargeting, 3DGS-based data generation, task representations, and capture geometry. Chapter 3 describes the pipeline formulations and assumptions. Chapter 4 reports the implementation and available evaluation. Chapter 5 examines measurement validity, failure cases, and the limits of generalisation. Chapter 6 summarises the supported findings and identifies the questions that require further experiments.

Глава 2

Literature Review

2.1 Preamble

Generating robot demonstrations from human recordings connects several research problems: recovering hand–object interaction, adapting that interaction to another embodiment, reconstructing the scene, and producing observations consistent with the resulting robot motion. The literature contains both specialised methods for these problems and integrated data-generation systems. The purpose of this review is to identify the representations and assumptions that can support the proposed ViKi workflow, and the measurements needed to assess their combination.

The chapter retains work on teleoperation, calibration, geometric fusion, and downstream learning because these areas establish requirements on the generated data even when they are not themselves the final application of the thesis. Teleoperation offers reference retargeting formulations; calibration and fusion determine the metric consistency of the source; learning systems define the observations and action labels that a usable dataset must supply. Their relevance is therefore methodological as well as comparative.

Section 2.2 examines human-motion representations and acquisition interfaces. Section 2.3 reviews retargeting objectives, temporal processing, and solvers. Section 2.4 discusses 3DGS-based rendering and demonstration generation. Sections 2.5 and 2.6 address task coordinates and capture geometry. Section 2.7 considers dataset construction and assessment. Section 2.8 positions ViKi relative to the closest systems, and Section 2.9 synthesises the design implications. The review is a narrative comparison of the cited publications and technical documentation, rather than evidence of an exhaustive absence of related work.

2.2 Representing the Human Hand

2.2.1 Articulated hand and body reconstruction

The representation of a demonstration determines which aspects of manipulation can be transferred. A wrist trajectory captures global hand motion, whereas finger articulation and hand–object geometry are needed to distinguish contacts that share a similar wrist pose. DexMV combines hand and object pose estimation with demonstration translation for dexterous robot learning [3]. OKAMI uses a single RGB-D video to derive a manipulation plan, retargeting body and hand motion with object-aware adaptation [4]. These works establish that human reconstruction and robot action generation are connected stages, rather than interchangeable outputs.

DexCanvas provides an instrumented reference for richer interaction capture [11]. Its acquisition system combines 22 infrared motion-capture cameras, two RGB-D cameras, and markers on the hand and objects. The observations are fitted to MANO and used in a real-to-simulation process that reconstructs contact-related forces through reinforcement learning. Its relevance to ViKi is twofold: articulated representations retain information needed by dexterous end-effectors, and contact annotations require evidence

beyond a visually plausible hand trajectory. Its marker-based acquisition also differs from the proposed markerless reconstruction route.

A fitted hand model can regularise noisy or partially observed landmarks, but it introduces assumptions about shape, articulation, and correspondence. Landmark error, mesh-fit error, and the accuracy of inferred contacts should therefore be evaluated separately. A parametric model used to create reference annotations need not be the model used by the evaluated perception pipeline. For ViKi, a useful representation preserves the available articulated observations while allowing target-specific reductions for simpler grippers. This avoids committing the entire source dataset to a single end-effector.

2.2.2 Compact arm and gripper representations

For a parallel-jaw target, a pose in $\mathbf{SE}(3)$ together with an opening command provides a compact action description. It can represent approach, transport, and approximate closing and release timing. It does not encode all finger contacts, in-hand manipulation, or the forces exerted by the demonstrator. Consequently, a scalar pinch signal can be an appropriate target abstraction without being a lossless representation of the original hand motion.

AnyTeleop supports different arms, hands, and camera configurations

within a vision-based teleoperation framework [12]. Open-TeleVision maps operator arm and hand motion to robot embodiments while providing immersive visual feedback [13]. Bunny-VisionPro combines bimanual dexterous teleoperation with haptic feedback and collision and singularity avoidance [14]. These systems motivate separating arm motion, finger correspondence, and embodiment-specific constraints instead of treating retargeting as a uniform rescaling of a human skeleton.

ACE further demonstrates that one collection interface can serve humanoid hands, arm–hand, arm–gripper, and quadruped–gripper configurations [15]. It combines a hand-facing camera with an exoskeleton to observe finger and wrist motion. Thus, teleoperation does not necessarily require a completely different operator interface for every robot. The distinction relevant to ViKi is whether a source demonstration can be recorded independently of a target robot and subsequently converted into several robot datasets.

The retargeting study of Xin et al. analyses the contribution of alternative hand-configuration objectives through ablations and manipulation experiments [16]. Its results support evaluating the choice and weighting of correspondences, rather than assuming that maximal similarity to every human joint is always desirable. Conclusions from a dexterous-hand ablation do not, by themselves,

determine the correct wrist weight for a different arm–gripper system. That weight depends on which task relationships the target must retain.

2.2.3 Egocentric capture and human–robot co-training

EgoMimic records egocentric human observations and 3D hand motion using Project Aria glasses and co-trains on human and robot demonstrations [17]. DexWild likewise uses human hand interactions collected across diverse environments, together with robot data, to improve policy transfer [18]. These are relevant alternatives to converting every human frame into a photorealistic image of a replacement robot. They show that human data can be useful through shared representations and co-training, although that route has different dataset requirements from explicit robot-action and robot-view synthesis.

Vision-Based Hand Shadowing provides a closer analytical video-to-robot workflow [19]. It uses an egocentric RGB-D camera, MediaPipe landmarks, depth lifting, damped-least-squares inverse kinematics, and a thumb–index-based gripper mapping for the SO-ARM101. Its reported degradation under occlusion makes visibility a relevant evaluation condition, not merely a reason to prefer more cameras. Multiview capture can add geometric evidence, but any improvement must be measured with controlled calibration, timing, and

view selection.

2.3 Retargeting Formulations

2.3.1 Correspondences, objectives, and confidence

Retargeting converts an observed human configuration or interaction into robot variables subject to a different kinematic structure. Position-based methods match selected points; vector-based methods match relative directions or lengths; task-space methods track a wrist, tool, or object relationship. AnyTeleop and the accompanying dex-retargeting toolkit illustrate several such formulations [12], [20]. Their common principle is to select geometric correspondences that the target can realise, while accounting for joint limits and temporal behaviour.

These objectives can be combined, but their weights have different meanings. A task weight expresses the importance of a relationship, while a perception weight expresses the reliability of its observation. A contact-dependent weight need not be a calibrated uncertainty estimate. In ViKi, this distinction matters when an occluded fingertip remains important to the task: increasing its task importance cannot compensate for a missing measurement. Confidence

weighting therefore needs an evaluation of its own effects on tracking, missing intervals, and failure cases.

The choice between wrist, fingertip, and object-relative residuals also depends on the target end-effector. For a parallel gripper, aperture and tool pose may be sufficient for a restricted pick-and-place task. For an Allegro-type hand, additional finger correspondences and grasp objectives are required. The representation should expose these differences rather than imply that one pinch mapping solves cross-morphology transfer in general.

2.3.2 Temporal objectives and sequential processing

Offline access to a complete demonstration permits temporal filtering, sliding-window optimisation, or optimisation over a complete robot trajectory. A sequential alternative reconstructs and smooths task-space motion, solves inverse kinematics frame by frame, and optionally filters joint commands. Because inverse kinematics is nonlinear and may switch solution branches, smooth Cartesian targets need not imply smooth joint motion. Conversely, filtering joint commands can change the end-effector path or invalidate previously checked clearances.

DexFlow combines hand-pose alignment with temporal consistency and

hand–object interaction refinement [21]. Its formulation includes sliding-window processing and motion prediction, and the authors discuss higher-order continuity in connection with regularisation. This supports examining temporal terms alongside pose alignment. It does not establish that any implementation using a smoothness penalty automatically satisfies a velocity, acceleration, or jerk bound.

A useful distinction is between a penalty and a constraint. Penalising large joint differences expresses a preference; enforcing a limit requires an explicit bound or an acceptance test on the resulting trajectory. Similarly, a robust residual may reduce the influence of an outlier but does not identify an unobserved contact or recover information lost during occlusion. The comparison of preprocessing and joint-space regularisation should therefore report both geometric tracking error and the motion derivatives affected by the chosen method.

2.3.3 Offline transfer and interaction-aware alternatives

HOWTransfer estimates temporally consistent hand motion, localises contact intervals, converts grasp intent into parallel-jaw grasp hypotheses, and edits the resulting trajectories [22]. Its hand-centric formulation is relevant

when reliable contact cues are available without complete explicit object-state tracking. It offers a distinct comparison to a pipeline that first reconstructs object poses and then imposes object-relative constraints.

ORION instead derives an object-centric manipulation plan from a human RGB-D video [23]. RoboWheel integrates human hand–object reconstruction, physical-plausibility refinement, cross-embodiment retargeting, and simulation augmentation [24]. These systems demonstrate that offline reconstruction, transfer, and data generation have already been studied in combination. The relevant comparison is between their assumptions and representations, not a division into integrated ViKi and otherwise isolated prior components.

For ViKi, hand-only, object-relative, and contact-aware retargeting can be compared using the same source motion and robot geometry. Contact-aware checking must distinguish permitted gripper–object contact from unintended penetration. It must also state which scene surfaces enter the collision model. A self-collision-free arm trajectory is not necessarily clear of the table, and imposing a recorded object trajectory does not show that the robot would physically cause that motion.

2.3.4 Inverse-kinematics libraries and trajectory optimisation

Pink provides a task-based differential inverse-kinematics interface built on Pinocchio [25]. Weighted tasks and supported limits or barriers can be assembled into local quadratic programs. This is useful for expressing end-effector targets and kinematic restrictions. However, a local program at one timestep and a batch problem coupling many timesteps have different variables and numerical properties. Choosing Pink does not, on its own, implement a complete trajectory-level objective.

The solver description should therefore distinguish local inverse kinematics, temporal coupling, and final verification. Joint limits, velocity limits, collision geometry, and numerical tolerances are properties of the configured application as well as the library. The resulting trajectory needs checks in the original geometry, particularly when linearised collision constraints or post-processing are used. This motivates retargeting-feasibility measurements without equating solver convergence with successful manipulation.

2.4 3DGS for Robot Learning

3DGS represents scene appearance using explicit Gaussian primitives with optimised position, covariance, opacity, and view-dependent colour. Kerbl et al. demonstrate high-quality novel-view synthesis with an efficient splatting renderer [5]. Its explicit spatial representation supports component editing and compositing, making it useful for generating observations beyond the original camera views. Robot actions, contact geometry, and time-varying object states must still come from another part of the system.

2.4.1 Augmenting recorded robot demonstrations

RoboSplat starts from a robot demonstration and multiview scene images, then edits reconstructed Gaussian components to generate additional training data [6]. Its six augmentation dimensions are object pose, object type, camera view, scene appearance, lighting, and embodiment. Thus, it already addresses embodiment variation as well as viewpoint and appearance changes. The distinction from the proposed ViKi acquisition stage is that its source motion comes from robot teleoperation. Its geometric editing operations should not be described as a general diffusion-based trajectory generator.

This example separates two forms of expansion. Re-rendering an existing trajectory changes the observation distribution; adapting object poses or robot embodiment may also require changing the action trajectory. Both can increase dataset coverage, but they create different validity requirements. A rendered robot must agree with its exported state, and a new geometric configuration must be checked for reachability and interaction consistency.

2.4.2 Simulation-based data generation and hybrid rendering

SplatSim uses Gaussian splatting for photorealistic observations coupled to a simulator [8]. Its experiments include demonstrations collected through simulated teleoperation or generated by motion planners, followed by policy training and real-world deployment. It is therefore relevant both as a rendering approach and as a synthetic-data workflow. It does not depend on recovering motion from a passive human recording.

RoboSimGS combines Gaussian scene appearance with interactive mesh assets and physics simulation [26]. Its pipeline includes inference of object properties and articulated structures using a multimodal model. The method is relevant to ViKi's separation of appearance and interaction geometry: photorealistic background reconstruction and usable collision assets are related

but different reconstruction products.

LEGS uses a mesh foreground for the robot and objects over a reconstructed 3DGS background, together with procedural motion primitives for humanoid loco-manipulation [27]. It also studies alignment between rendered appearance and the deployment camera. Its separation of motion and appearance allows demonstrations to be re-rendered under changed scene conditions. The human-motion source differs from ViKi's proposal, but the hybrid rendering architecture and appearance calibration remain directly relevant.

These systems establish that a 3DGS-based pipeline need not render every component as Gaussians. A mesh foreground can supply articulation and depth while a splat background supplies appearance. Such a combination still requires metric alignment, compatible camera models, and correct visibility ordering. A visually detailed background does not compensate for a misplaced robot or an incorrect object state.

2.4.3 Planning demonstrations from reconstructed scenes

GaussianFactory generates demonstrations from a video scan by reconstructing editable 3DGS assets and planning task motion geometrically [9]. Grasp formation uses a pretrained contact model, and the work includes downstream

policy evaluation. Its input is scene structure rather than a human manipulation trajectory. This offers an alternative motion source for dataset generation, not evidence that procedurally generated motion is inherently inferior. A comparison with demonstration transfer should consider the tasks covered, preparation effort, and the interaction information that must be specified or inferred in each case.

2.4.4 Human demonstrations and novel-view synthesis

Tool-as-Interface is a particularly relevant precedent [28]. It uses two RGB views, MASt3R reconstruction, and Gaussian splatting to augment human tool-use observations across viewpoints. Tool poses provide task-space action labels, while segmentation removes embodiment-specific regions from human and robot observations. This combines human demonstration reuse with 3D-aware visual synthesis without requiring the complete human hand to be reconstructed. Its shared-tool representation differs from explicitly rendering target robot embodiments, but its existence rules out treating human-to-robot transfer and Gaussian view augmentation as unrelated lines of work.

Real2Render2Real uses a smartphone object scan and a monocular human

manipulation video to recover object-part motion and generate robot trajectories [2]. It uses 3DGS for asset construction and tracking, then converts reconstructed assets to textured meshes for rendering in IsaacLab with dynamics disabled. The project supports retargeting across robot embodiments and variation in camera and scene conditions. Its sensing and final rendering differ from ViKi's proposed multiview RGB-D route, but its reuse of human-demonstrated object motion is a close methodological precedent.

RoboPaint more directly combines multiview RGB-D human capture, retargeting, and 3DGS-based visual synthesis [7]. It records glove kinematics and tactile signals alongside camera observations, uses wrist markers, and retargets motion with kinematic and tactile objectives. Its hybrid renderer combines a static 3DGS background with mesh robots and moving objects, supporting novel views and scene edits. The principal acquisition distinction proposed for ViKi is recovery from camera observations without worn hand markers or instrumented gloves. That distinction is a design choice to evaluate, rather than proof that the broad integration is new.

2.4.5 Reconstruction boundaries and evaluation

The reviewed methods use direct splat rendering, hybrid rendering, or 3DGS-to-mesh conversion for different purposes. These choices affect how objects are segmented, how trajectories drive the scene, and which surfaces are observable from new cameras. A manipulation recording also contains a moving demonstrator and possibly moving objects; a static reconstruction cannot simply be assumed to reproduce all of them consistently over time.

ViKi therefore needs an explicit decomposition into the background, moving object assets, and replacement robot, together with a method for removing the human from generated observations. Reconstruction quality can be measured against held-out views of the same scene state. Replacement-robot compositing requires additional checks on depth, boundaries, temporal consistency, and agreement with exported states. A human image is not a paired pixel-level reference for an image containing a different robot. These distinctions keep visual fidelity separate from the correctness of the generated action labels.

2.5 Task Representation

2.5.1 World, object, and tool coordinates

A trajectory expressed in the capture workspace is straightforward to inspect, but replaying it after an object has moved may miss the intended interaction. Object-centric and tool-centric representations instead preserve selected relations between the end-effector and the task. ORION extracts an object-centric plan, and SPOT represents task motion using an object-pose trajectory relative to a target [23], [29]. Their relevance is the separation of task structure from the particular joint trajectory used to realise it.

OKAMI occupies a related position: object-aware adaptation modifies recovered human motion for new object locations [4]. Vi-RoIL expresses wrist trajectories and key-frame hand poses in object coordinates after monocular hand–object reconstruction with a wrist-mounted marker cube [30]. Its reported qualitative simulation results support discussion of coordinate consistency, while providing a different evidence level from a quantitative real-robot manipulation benchmark.

Tool-as-Interface uses the tool as a shared physical interface between the human and robot [28]. This is particularly useful when the tool can be rigidly

attached or grasped with a known transform. It also illustrates the limits of a universal wrist representation: task-relevant motion may be more naturally specified at the tool than at an anatomical hand joint.

2.5.2 Flow representations and non-rigid interaction

NovaFlow generates task videos from descriptions, extracts actionable object flow, and uses that flow for robot planning [31]. For rigid objects, it derives relative poses; for deformable objects, it uses flow as an objective for model-based planning. Its motion source is generated video rather than a recorded human demonstration. Nevertheless, it is relevant to the representation choice because it shows how a flow-based interface can extend beyond a single rigid-pose trajectory.

A rigid-object ViKi pipeline should therefore identify its object assumptions without implying that all demonstration-driven transfer requires rigidity. Cloth, fluids, and changing topology require a different motion and interaction model. Even for rigid objects, symmetry and occlusion can make orientation ambiguous. A stable-looking object track may still impose an incorrect relative grasp when used downstream.

2.5.3 Implications for reusable demonstrations

Object-relative motion can be stored alongside world-frame observations, as considered in Section 3.4. Retaining both allows comparison of direct hand tracking with interaction-aware retargeting and preserves the original evidence when an object estimate fails. Object-relative coordinates do not create invariance to arbitrary scene changes: the new layout must still admit the interaction and a feasible robot path.

The relevant preservation target is task-dependent hand–object motion and interaction timing. A robot with a different reach or end-effector may need a different approach or grasp. Consequently, cross-embodiment reuse should report which relationships were preserved, which were adapted, and which source episodes could not be transferred under the declared constraints.

2.6 Multiview RGB-D Capture Geometry

2.6.1 Reference data and camera configuration

HO-Cap provides eight RealSense views and a HoloLens recording, together with semi-automatic hand and object annotations [10]. It is relevant both as a source of manipulation recordings and as an external reference for perception

assessment. The annotations are not an independent optical-motion- capture measurement of ViKi's own rig. Dataset-specific performance and local capture accuracy remain separate quantities.

Camera count, baseline, overlap, viewing direction, and timing jointly affect what can be reconstructed. Additional views may reveal occluded fingers or object surfaces, but inconsistent calibration can also introduce conflicting constraints. A camera subset selected on the evaluation frames measures a development choice rather than held-out generalisation. This motivates controlled comparisons of camera configurations without making two cameras a defining property of the proposed method.

2.6.2 Fiducials and multiview registration

ChArUco combines marker identification with chessboard-corner localisation and supports calibration from partially visible boards [32]. The appropriate target depends on visibility, workspace size, and whether intrinsics, extrinsics, or a workspace anchor are required. A small image reprojection error is one useful diagnostic; it does not alone establish metric agreement between depth clouds or accurate camera-to-robot registration.

Romeo et al. compare two- and three-dimensional calibration procedures

for Azure Kinect data [33]. Their reported correspondence errors differ between colour and infrared reconstructions and between static and dynamic experiments. Darwish et al. calibrate a five-Azure-Kinect system with a ChArUco-coded cube and combine reprojection with point-to-plane and patch-to-plane terms [34]. Both studies motivate evaluating spatial consistency in the reconstructed geometry, rather than treating an image-space residual as the only calibration criterion.

Kim et al. initialise multiview registration using body tracking and refine it through feature matching and mismatch rejection [35]. Their feature inlier ratios describe the refinement procedure, not directly the absolute accuracy of a manipulation trajectory. Bu and Lee similarly combine body-tracking-based coarse alignment with ICP refinement [36]. Such approaches offer markerless calibration alternatives, but depend on initial alignment and sufficient geometric information for refinement. They remain relevant to camera placement even if ViKi uses a fiducial workspace anchor.

2.6.3 Sensor characterisation and metric accuracy

Tölgyessy et al. evaluate Azure Kinect precision, systematic error, warm-up, material response, and depth artifacts [37]. Kurillo et al. compare Azure

Kinect and Kinect v2 using reference geometry and analyse variation with range and image position [38]. These studies show why depth quality should be characterised under the actual sensing conditions: central planar surfaces, object boundaries, reflective materials, and partially occluded hands do not present the same estimation problem.

The RealSense D400 documentation likewise distinguishes depth accuracy, fill rate, spatial noise, and temporal noise under stated test conditions [39]. These are useful specifications for choosing and testing capture settings, but they are not interchangeable with multiview registration error, 3D hand-joint error, or robot TCP error. Nor does an upper specification bound establish a lower error floor for all measurements. ViKi's metric claims need controls on the relevant rig and clearly defined reference quantities.

2.6.4 Depth lifting, triangulation, and fusion

Depth lifting estimates a surface location from an image coordinate and a depth sample. Multiview triangulation instead combines viewing rays, optionally with depth constraints. For hand tracking, the visible skin surface and anatomical joint centre are not identical. For scene reconstruction, missing depth and mixed pixels at boundaries can affect geometry and subsequent

occlusion rendering. These distinctions motivate separating sparse landmark fusion from dense surface reconstruction.

The ray-casting fusion described in U.S. Patent 10,818,071 aggregates signed-distance information from several depth images and includes weighting related to view geometry and visibility [40]. It is relevant as a concrete geometric fusion design, rather than comparative evidence that a particular weighting law is optimal for hand landmarks.

Zeng et al. learn confidence-dependent RGB-D feature fusion for surface-normal estimation [41]. Their contribution concerns weighting depth features before combining them with colour features in a single-image prediction network. It supports the broader treatment of depth reliability, but is not a direct validation of multi-camera joint averaging. HRBF-Fusion uses continuous implicit surfaces for camera tracking and derives curvature and confidence information for RGB-D fusion [42]. It provides a geometric alternative concerned with reconstruction quality and tracking robustness.

These approaches justify considering uncertainty and local geometry when combining observations. For ViKi, the reliability signal still needs to be defined and tested: detector confidence, depth validity, reprojection agreement, and interpolated samples represent different kinds of evidence. Filling a trajectory

gap increases completeness without adding an observation, so coverage should be reported alongside reconstruction error.

2.6.5 Exposure synchronisation and software alignment

Azure Kinect supports wired multi-device synchronisation and configurable exposure delays; its documentation also addresses interference between depth emitters [43]. RealSense documentation describes its own capture and synchronisation interfaces [39]. The applicable trigger modes and timestamp domains must be established for the actual device combination, rather than inferred from the fact that both families expose timestamps.

RidgeRun's RR Syncer documentation illustrates a separate software layer: aligning video buffers by presentation timestamps, applying a synchronisation window, and dropping unmatched buffers [44]. Buffer alignment is useful for managing streams, but does not by itself establish simultaneous exposure. Similarly, a host monotonic clock measures arrival or application timing unless a mapping to sensor acquisition time has been established.

For heterogeneous capture, the defensible requirement is a stated synchronisation procedure with measured residual offsets and dropped-frame behaviour. The literature does not justify declaring hardware coordination between camera

families impossible or treating host-side nearest-timestamp pairing as the only available mechanism. Timing accuracy also matters to 3DGS reconstruction when moving hands and objects are observed from several cameras.

2.6.6 Camera-to-robot registration

The generated robot must share a metric frame with the reconstructed scene. A workspace anchor can connect fixed cameras to a calibrated robot-base pose; hand–eye calibration is another relevant tool when suitable camera and robot motions are available. The comparative study by Enebuse et al. finds that hand–eye algorithm behaviour depends on translation and rotation noise and on the motion range used for calibration [45]. Its relevance is the need to match and test the procedure under the intended conditions, rather than to declare one solver universally preferable.

For dataset generation without a physical target robot, the chosen base pose is a virtual placement whose transform and reachability still need to be recorded. For later deployment, an independently established real camera-to-robot transform is required. Neither case is validated solely by a plausible overlay of the rendered robot on the captured scene.

2.7 Dataset Generation and Downstream Requirements

2.7.1 Demonstration reuse and episode structure

MimicGen adapts segments of a limited set of robot demonstrations to generate additional demonstrations under new scene configurations [1]. Together with the human-source pipelines reviewed above, it establishes reuse as a general data-generation strategy. The important distinctions are the source of task motion, the assumptions used for adaptation, and the checks applied to the generated episodes.

For ViKi, a learning episode needs temporally consistent observations, robot states, and actions, with explicit units and coordinate conventions. The rendered camera pose and scene state must correspond to the exported motion. Changing the robot can change action dimension and limits; changing a camera changes observations even when the joint trajectory is unchanged. A trajectory bundle and a rendered observation–action dataset are therefore distinct outputs. Compatibility with a format such as LeRobot concerns organisation and interfaces, not certification of physical task success.

2.7.2 Learning architectures as downstream consumers

ACT, Diffusion Policy, and vision-language-action models motivate retaining synchronised observations and action sequences, even though training them is outside the present evaluation. OpenVLA-OFT studies action decoding, continuous action representations, action chunking, and fine-tuning objectives [46]. Its relevance is that action representation and temporal organisation affect how a dataset can be consumed. Its policy results are not evidence that a new synthetic dataset will be useful without a corresponding learning experiment.

This downstream perspective also favours retaining robot identity, camera calibration, source episode, retargeting parameters, and editing provenance. Those fields allow consumers to distinguish new observations of an existing action sequence from genuinely adapted motion, and make failures traceable to the relevant pipeline stage.

2.7.3 Curation and evidence of quality

Demo-SCORE filters heterogeneous demonstration data using a classifier trained and cross-validated on successful and unsuccessful policy rollouts [47].

It is relevant because increasing demonstration count can retain unreliable strategies as well as useful variation. Its quality signal, however, depends on online experience. A pipeline without such rollouts cannot claim equivalent success-based screening from trajectory smoothness alone.

ViKi can assess reconstruction error, geometric feasibility, rendering consistency, and coverage before policy training. These are useful curation criteria with a narrower interpretation than demonstrated manipulation success. Geometric acceptance should specify which constraints were tested, and rejected episodes should remain visible in the generation statistics. A visually plausible sequence or a valid file schema cannot substitute for that information.

2.7.4 Factorial expansion, diversity, and generation effort

With M embodiments, K camera viewpoints, and P appearance configurations, a source episode can define up to MKP candidate configurations. The actual yield depends on reachability, grasp compatibility, scene observability, and the stated acceptance checks. Multiple views of one motion improve observation coverage without becoming independent manipulation strategies. Appearance changes may increase visual diversity without changing task geometry.

Consequently, dataset characterisation should report requested and accepted

combinations, failure categories, camera coverage, trajectory variation, and processing time. Feature-space distances can provide supplementary measures of visual variation when the encoder and sampling procedure are fixed. They do not alone establish learning benefit or coverage of additional skills. Likewise, a cost comparison with teleoperation needs measured operator, calibration, annotation, scanning, computation, and rejection costs. Savings cannot be inferred from the nominal expansion factor alone.

2.8 Positioning of the Proposed System

The closest comparisons depend on the stage under consideration. Instrumented capture systems help assess what interaction information can be recovered; teleoperation systems supply retargeting alternatives; geometric reconstruction and rendering systems define visual-generation choices. Table 2.1 compares representative integrated workflows by their motion source and visual representation. It avoids equating a planned ViKi capability with a capability already evaluated by another system.

TABLE 2.1
Motion Sources, Representations, and Visual Generation in Related Work

System	Motion source and representation	Visual generation and relation to ViKi
HOWTransfer [22]	Human hand motion and contact intervals; grasp hypotheses and trajectory editing.	Relevant offline transfer alternative; not the same objective as synthesising robot-view datasets.
Vi-RoIL [30]	Monocular human motion with marker-assisted calibration; object-relative representation.	Relevant geometric front-end; reported evaluation is qualitative simulation.
Tool-as-Interface [28]	Two-view human RGB recordings; tool-centric actions.	Gaussian novel-view augmentation and embodiment masking, rather than replacement-robot rendering.
Real2Render2Real [2]	Human video and object-part motion; robot-specific grasping and IK.	3DGS-to-mesh assets and synthetic rendering, including embodiment reuse.
RoboPaint [7]	Multiview RGB-D, glove kinematics, tactile data, and wrist markers.	Human-to-robot retargeting with a 3DGS background and mesh foreground.
RoboWheel [24]	Human interaction reconstruction and cross-embodiment retargeting.	Simulation augmentation; integrated precedent for human-source data generation.
RoboSplat [6]	Robot demonstration with geometric augmentation.	Editable Gaussians for embodiment, viewpoint, and scene variation.
SplatSim [8]	Simulated teleoperation or planned demonstrations.	Gaussian observations coupled to simulation; includes policy-transfer evaluation.
GaussianFactory [9]	Grasp and trajectory planning from a scene scan.	Editable 3DGS scene; planning supplies the motion.
LEGS / RoboSimGS [26], [27]	Procedural motion or simulated interaction.	Hybrid Gaussian appearance and mesh foreground; different motion-generation assumptions.

System	Motion source and representation	Visual generation and relation to ViKi
ViKi (proposed)	Markerless multiview RGB-D hand–object motion and embodiment-specific retargeting.	Proposed robot-view rendering and controlled expansion, assessed through stage and integration checks.

RoboPaint already combines the broad components in the original motivation, while Tool-as-Interface and Real2Render2Real provide additional close precedents for human-source motion and visual augmentation. ViKi is therefore positioned as an investigation of a particular markerless multiview RGB-D route and its operating limits. The intended contribution is to implement and characterise that route across declared capture, embodiment, and rendering conditions, including failures at their interfaces. The camera setup and target robot set are experimental variables, rather than grounds for an unsupported claim of first integration.

2.9 Conclusion of the Review

The surveyed literature supports the proposed direction while making its assumptions and evidence requirements explicit. The following implications connect the review to the methodology:

1. Retain a reusable human representation. Articulated hand observations and hand–object relationships can support different target abstractions. A wrist pose and scalar aperture are suitable for some grippers, while dexterous hands require richer transfer.

2. Compare retargeting formulations under matched conditions. Evaluate hand-only and interaction-aware objectives using the same recordings and robot placements. Report tracking error, motion regularity, and constraint violations separately.

3. Treat geometry and appearance as connected representations. Specify the static background, dynamic objects, human removal, and replacement robot, together with their metric alignment and visibility handling. Photorealistic appearance does not establish contact validity.

4. Measure capture quality on the relevant setup. Calibration, depth validity, camera placement, and timing affect both perception and novel-view synthesis. External reference data and local rig measurements answer different questions.

5. Validate the interfaces between stages. Alongside independent stage metrics, check coordinate frames, time alignment, object state, and agreement

between rendered robot motion and exported actions.

6. Characterise accepted expansion rather than nominal size. Record configuration coverage, rejected cases, and preparation and generation effort. Keep those findings distinct from physical task success and policy-learning performance.

These choices define the proposed dataset-generation study without presupposing its outcome. Chapter 3 develops the relevant formulations, while implementation and experimental evidence determine which capabilities can ultimately be claimed.

Глава 3

Design and Methodology

3.1 System Overview and Implementation Boundary

ViKi is designed as an offline pipeline that separates human-demonstration capture, motion reconstruction, robot-specific retargeting, and synthetic observation generation. Recording a human episode does not require the demonstrator to control a robot. Subsequent processing may use the complete episode, rather than meeting the latency requirements of teleoperation. This permits temporal fitting and trajectory-level optimisation, while making their processing cost and use of future observations explicit.

The target workflow has two branches after capture. The motion branch

estimates hand and object trajectories and derives robot joint and gripper commands. The appearance branch reconstructs reusable scene assets and renders the resulting robot motion from selected cameras. Their outputs must agree on coordinate frames, timestamps, object identities, and robot state. Figure 3.1 shows this design.

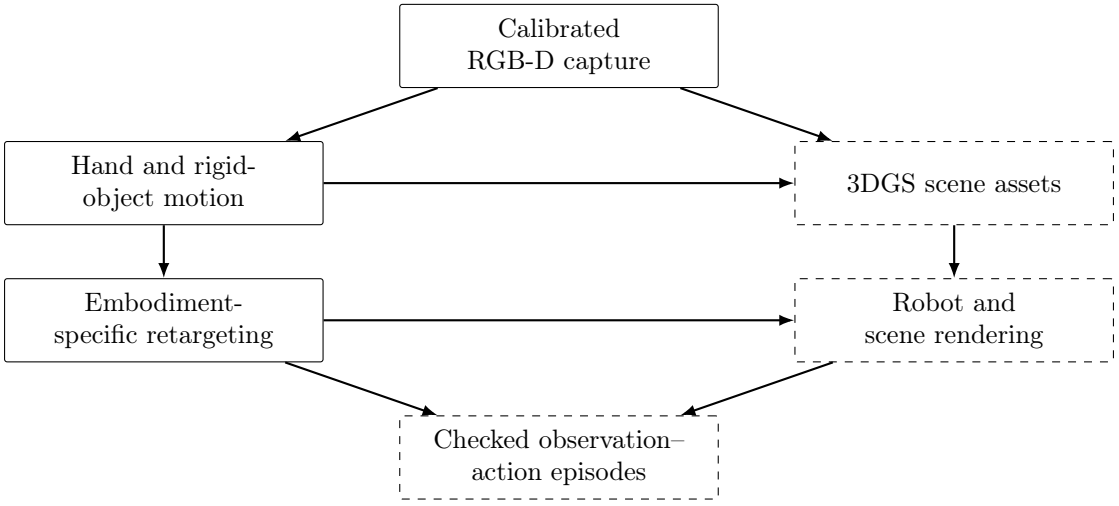


Fig. 3.1. Target dataset-generation workflow. Solid boxes have corresponding processing code in the inspected revision; dashed boxes denote the proposed 3DGS and synthetic-dataset extension. An implemented stage does not imply that its accuracy or downstream task performance has been established.

The implementation description is tied to repository revision 0fdbc02 of 7 October 2026.[1] At that revision, capture, hand processing, rigid-object modelling, trajectory-level inverse kinematics (IK), object-relative diagnostics, and trajectory export have code implementations. An experimental cube-specific grasp planner also exists. The 3DGS reconstruction and robot-compositing

[1]Inspected source: ViKi repository at the pinned revision.

stages described below are proposed extensions; the existing point-cloud viewer and image utilities do not implement them. Physical robot replay remains a stub, and the optional LeRobot writer is partial.

Stage outputs are stored separately so that a failed downstream stage does not require recapturing the demonstration. The principal current artifacts are raw RGB-D recordings, per-camera landmarks in rec.npz, prepared hand trajectories in cln.npz, rigid-object models and tracks in object_models.npz, and robot plans in plan.h5. Object-relative tracks are an additional artifact computed after IK. The proposed renderer consumes these estimates; it does not turn them into independently measured robot demonstrations.

3.2 Capture Geometry and Coordinate Frames

3.2.1 Hardware and workspace

The local reference configuration uses two Azure Kinect RGB-D cameras viewing a bounded tabletop workspace. Camera placement should provide overlapping coverage of the hand and manipulated objects, while avoiding nearly parallel viewing rays in the interaction region. More views can reduce particular occlusions, but their benefit depends on placement, calibration,

detector errors, and depth quality. Neither the number of cameras nor nominal sensor specifications establish reconstruction accuracy [36], [37], [43].

The acquisition layer also contains a RealSense backend. Supporting a backend is distinct from validating a particular mixed-camera rig. Every experiment must record the devices, colour and depth modes, exposure settings, frame rate, synchronization mode, and calibration used. Camera motion invalidates the corresponding fixed-rig calibration. The robot is introduced as a kinematic model during offline processing; a physical robot is not required for human capture.

3.2.2 Calibration and frame convention

Let $\mathbf{T}_B^A$ map coordinates from frame $\boldsymbol{B}$ to frame $\boldsymbol{A}$. The implementation distinguishes a camera-rig frame $\boldsymbol{G}$, a ChArUco calibration/workspace frame $\boldsymbol{C}$, and the base frame $\boldsymbol{B}_m$ of target robot $\boldsymbol{m}$. The rig frame is tied to a reference camera. Inter-camera extrinsics and the workspace anchor are separate calibration quantities, rather than two names for the same world frame.

For a point observed by camera k, the complete transformation is

$$\begin{bmatrix} \mathbf{x}^C \\ \\ 1 \end{bmatrix} = \mathbf{T}_G^C \, \mathbf{T}_k^G \begin{bmatrix} \mathbf{x}^k \\ \\ 1 \end{bmatrix}. \tag{3.1}$$

The camera geometry includes intrinsics and distortion or factory ray models, together with the relationship between colour and depth sensors. ChArUco observations provide constraints for the rig solve and workspace anchor [32]. In the Azure Kinect path, factory-calibrated rays are used to place colour observations in an ideal pinhole plane for the multi-pose solve. This avoids treating raw distorted pixels as if they already followed that ideal model.

The saved calibration must retain the board dimensions and marker dictionary actually used. A numerically successful calibration solve or an automatic validation verdict is insufficient evidence of metric accuracy throughout the task volume. Reprojection residuals, cross-view surface agreement, and an independent dimension or distance check answer different questions and should be reported separately.

For retargeting, the configured robot-base pose is $\mathbf{T}_{B_m}^C$, giving $\mathbf{T}_H^{B_m} = (\mathbf{T}_{B_m}^C)^{-1}\mathbf{T}_H^C$ for a human target frame H. The current code accepts a full base translation and orientation from configuration. It does not estimate this pose

through a physical hand–eye calibration procedure. The transform therefore defines a virtual placement for geometric evaluation. Use on hardware would require an independently verified registration of the actual robot base. No automatic workspace rescaling or recentering is part of this transformation.

3.2.3 Synchronisation

The recorder stores both device timestamps and host timestamps. Host time is derived from a monotonic clock,

$$\tau_{\text{host},\mu\text{s}} = \left\lfloor \frac{t_{\text{monotonic,ns}}}{1000} \right\rfloor, \tag{3.2}$$

and is used to select nearby buffered frames around a recording tick. The host-time acceptance window is a stale-frame guard; it does not certify simultaneous exposure.

For a configured hardware-synchronised Kinect rig, the implementation checks master/subordinate roles and timestamp-offset stability before recording. It then matches subordinate device timestamps to the master, allowing for the measured offset. The inspected revision uses a $500\,\mu\text{s}$ residual tolerance for this device-timestamp check. That configured tolerance is not a measured

bound on optical exposure alignment or on the reconstruction error of an episode. Other acquisition configurations use host-based grouping unless a verified device-time relationship is available.

Offline multiview reconstruction groups observations by recorded synchronised frame index. Different cameras' host timestamps need not be identical within that group. An optional image-landmark time-alignment step is a distinct processing choice and must be recorded in the profile. Local synchronization assessment should report device-time residuals, host-arrival offsets, dropped groups, and drift over the recording. An external benchmark cannot validate the timing of the local camera rig.

3.3 Hand Reconstruction and Temporal Preparation

3.3.1 Image observations and depth evidence

The working MediaPipe backend supplies 21 hand landmarks per detected hand in each colour image. Each observation contains image coordinates, a detector score, and, where available, a local depth measurement and depth spread. Detector confidence is evidence used by the estimator, not a calibrated probability of 3D correctness. Depth lifting uses the camera's colour–depth

geometry and converts measurements to metres.

Per-camera observations and depth-lifted trajectories are retained before cross-camera fusion. This permits comparison of triangulation with the legacy depth-based path without rerunning the detector. Missing detections and invalid depth are recorded explicitly rather than replaced by zero coordinates.

3.3.2 Calibrated multiview triangulation

The named multiview profiles use triangulation as their primary fusion method. For each landmark and frame, camera pairs provide direct linear triangulation hypotheses. A hypothesis must lie in front of both cameras and satisfy a minimum angle between its viewing rays. Candidate hypotheses are ranked by reprojection-inlier count, then inlier detector support and reprojection error. The best supported hypothesis is refined over its inlier views.

For the angular-residual mode, let s_k be the detector score, f_k the camera focal scale, d_k a valid depth observation, and $z_k(\mathbf{x})$ the predicted camera-axis

depth. The residual for view k contains

$$\mathbf{r}_k(\mathbf{x}) = \begin{bmatrix} \sqrt{s_k}\,[\pi_k(\mathbf{x}) - \mathbf{u}_k]/f_k \\ \sqrt{\lambda_d w_k^d}\,[z_k(\mathbf{x}) - d_k - \delta_i]/\max(z_k(\mathbf{x}), \varepsilon) \end{bmatrix}, \tag{3.3}$$

$$\hat{\mathbf{x}}_{i,t} = \arg\min_{\mathbf{x}} \sum_{k \in \mathcal{K}_{i,t}^{\text{in}}} \sum_a \rho([\mathbf{r}_k(\mathbf{x})]_a)\,.$$

Here π_k projects into the undistorted image convention used by the solver. The depth row is omitted when depth is invalid. Its weight decreases with local depth spread; δ_i is a configurable surface-to-joint depth offset. The robust loss acts on scalar residual components. The legacy pixel-residual mode has a different scaling and is preserved as a separate profile choice.

Triangulation does not remove range-dependent uncertainty. Small ray intersection angles, inaccurate calibration, motion between views, and consistent image-localisation errors can all produce an inaccurate 3D point. Its geometric quality score is

$$q_{i,t} = \frac{n_{\text{in}}}{n_{\text{usable}}} \left[1 - \frac{\bar{e}}{2\varepsilon_{\text{in}}}\right]_0^1 \left[\frac{\theta_{\text{best}}}{\theta_{\text{ref}}}\right]_0^1, \tag{3.4}$$

where $[\cdot]_0^1$ denotes clipping, $\bar{e}$ is mean inlier reprojection error, and θ_{best} is the largest inlier-pair ray angle. Profiles may additionally multiply this quantity by mean inlier detector score. Quality depends on the chosen gates and is not

an independent accuracy measurement. With fewer than two usable views, this triangulation path leaves a gap.

3.3.3 Legacy fusion, interpolation, and articulated fitting

The alternative xyz_mean path combines finite depth-lifted points on a common time grid:

$$\hat{\mathbf{x}}_{i,t} = \frac{\sum_k w_{i,t}^k \mathbf{x}_{i,t}^k}{\sum_k w_{i,t}^k}. \tag{3.5}$$

Its current weights come from the supplied per-camera confidence arrays. The implementation does not compute the previously proposed product of detector confidence, depth-sensor confidence, inverse squared range, and surface-incidence angle. This legacy mode can operate on single-view depth evidence, but it is not a per-landmark automatic fallback inside the strict named triangulation profiles. A custom run that falls back because a triangulation artifact is missing must be identified as such.

After fusion, missing coordinates may be interpolated and the landmark sequence smoothed with a Savitzky–Golay filter. These are operations on landmark coordinates, not cubic interpolation on $\mathbf{SE}(3)$. The inspected default profile, stable-fused-hand-v4, uses linear fused-gap interpolation, permits edge

holding and unbounded gap length, and expresses several spatial and temporal settings in angular units or milliseconds. Consequently, a finite prepared coordinate can be temporally filled rather than directly supported by camera observations. Observed masks, gap lengths, and fitted-pose support must accompany accuracy and coverage statistics.

An additional articulated-hand stage fits a constrained landmark/capsule model to the fused sequence. The stable landmark-only fitting path does not use a MANO surface model or dense depth fitting. Its outputs are saved alongside the original prepared landmarks, allowing fitted and landmark poses to be compared. Kinematic regularity of this fitted hand is useful internal evidence, but it is not independent proof that the recovered hand motion is correct.

3.3.4 Palm frame, interaction point, and gripper signal

The landmark-derived palm frame uses wrist and metacarpophalangeal (MCP) landmarks. Its longitudinal axis points from the wrist to the middle MCP; the normal is the normalised cross product of that axis with the index-to-pinky MCP vector. The remaining axis completes a right-handed frame. Degenerate or missing geometry requires a validity decision; an identity-

orientation placeholder is not an observed palm rotation.

The robot target need not be located at the wrist. The current pinch-centre option uses the midpoint of the thumb and index fingertips for target position, while orientation can come from the selected palm-pose source. Thus a run can combine a landmark-derived interaction point with an articulated-fit orientation. Both choices must be retained in provenance.

For parallel-jaw transfer, the continuous gripper decoder uses a dimensionless pinch ratio rather than treating fingertip distance as the robot's metric aperture:

$$
\begin{aligned}
r_t &= \frac{\|\mathbf{x}_{\text{thumb},t} - \mathbf{x}_{\text{index},t}\|}{\|\mathbf{x}_{\text{wrist},t} - \mathbf{x}_{\text{middleMCP},t}\|}, \\
g_t^{\text{raw}} &= \left[\frac{r_t - r_{\text{closed}}}{r_{\text{open}} - r_{\text{closed}}}\right]_0^1, \qquad g_t = g_{t-1} + \alpha_g(g_t^{\text{raw}} - g_{t-1}).
\end{aligned} \tag{3.6}
$$

Here zero denotes closed and one fully open. Thresholds and smoothing are decoder parameters; the first valid sample initialises the filter, and a missing sample holds the previous command. A binary hysteresis decoder is also available. These mappings are design choices, rather than anatomical equivalences between a human hand and a robot gripper. As in other human-to-robot interfaces, the correspondence must match the target end-effector

[13], [15]. A scalar opening signal does not specify a complete dexterous-hand configuration or contact force.

3.4 Object Motion and Relative Representation

3.4.1 Segmentation and rigid modelling

The implemented scene branch uses prompted SAM 2.1 video segmentation [48]. Prompts associate masks with persistent object identities and roles. An optional empty-background depth plate supports automatic prompt proposals from foreground components, but these proposals do not establish which object is manipulated or which surfaces define the task. Prompt correction and role assignment are recorded inputs.

Calibrated RGB-D lifting converts masks into semantic point observations. The object modeller builds a temporally supported canonical core, tracks it with robust trimmed nearest-neighbour alignment and weighted rigid fitting, then adds supported shell points and refines the track. Operator-proximity flags suppress shell growth during possible contact, reducing the risk of incorporating hand points into the object model. They indicate geometric proximity, not measured contact or a stable grasp.

The output contains object poses, confidence, support, rotation-information diagnostics, and validity. Partial visibility and object symmetry may leave rotation poorly constrained even when translation is stable. These artifacts are geometric models and motion estimates; they are not yet textured 3DGS assets and do not automatically alter the hand trajectory.

3.4.2 Relative tracks and proposed transfer

For a saved human frame H and object O, the object-relative pose is

$$\mathbf{T}_{H,t}^{O} = (\mathbf{T}_{O,t}^{C})^{-1}\mathbf{T}_{H,t}^{C}. \tag{3.7}$$

The same construction yields object-relative target and achieved robot tool poses. The current object_relative.npz artifact is computed after object modelling and IK. It permits analysis of relative motion and interaction error; it is not evidence that general object-relative retargeting drove the plan. Its saved human frame must also be distinguished from a pinch-centre robot target when those positions differ.

A possible object-relative target construction for embodiment m is

$$\mathbf{T}_{E,t}^{B_m,\mathrm{des}} = \mathbf{T}_{O',t}^{B_m}\,\mathbf{T}_{H,t}^{O}\,\mathbf{T}_{E,m}^{H}, \tag{3.8}$$

where O' is a specified target object trajectory and $\mathbf{T}_{E,m}^{H}$ is a declared human-to-tool correspondence. This equation defines a proposed transfer option, not the current general-purpose IK input. A constant correspondence is only an approximation for changing human grasps and different gripper geometries. Altering object placement may also require changing approach, support, and release conditions. Object-relative representations are motivated by their ability to express task geometry [23], [29]; representation alone does not make a transferred interaction feasible.

3.4.3 Implemented cube-specific pilot

The inspected revision includes an experimental object-aware planner for a cube and Robotiq 2F-85. It requires one explicitly identified manipulated object, supported canonical geometry with suitable side planes, and sufficient motion/contact evidence. Missing or ambiguous prerequisites cause an error rather than a silent hand-only fallback.

This planner separates approach, closure, carry, release, and retreat. During carry it follows object translation with a fixed pickup offset. Jaw orientation is aligned with supported cube geometry and held through the carry, rather than trusting uncertain framewise cube rotation. The opening command is derived from the modelled pad separation and the specified cube width. This is a restricted geometric grasp hypothesis. It intentionally modifies parts of the measured human motion, and neither its closure command nor its imposed object path demonstrates force closure or successful manipulation. Hand-only and cube-aware plans must therefore be evaluated as distinct methods.

3.5 Offline Trajectory Retargeting

3.5.1 Robot model and target construction

Each target embodiment specifies an arm model, joint ordering and limits, base placement, reference posture, gripper, mounting adapter, and human-to-tool correspondence. The current arm registry contains UR3, UR5, UR10, UR5e, UR10e, and iiwa14 descriptions. A registry entry is not a cross-embodiment evaluation result. Panda, Gen3, and xArm7 remain candidate

additions until their models and end-effectors are integrated and tested.

The working Robotiq 2F-85 assembly uses the actual gripper geometry. Its positional task point is the midpoint of the two modelled contact-pad centres, and orientation is taken from the designated tool frame. Because the pad positions vary with gripper opening, the positional task is not a fixed flange offset. Let $\mathbf{q}_t \in \mathbb{R}^{n_m}$ contain the arm joints and let g_t determine the gripper's drive and mimic joints. Forward kinematics then provides $\mathbf{p}_m(\mathbf{q}_t, g_t)$ and $\mathbf{R}_m(\mathbf{q}_t, g_t)$. Gripper rate profiling precedes the arm optimisation; gripper joints are known framewise inputs rather than additional arm decision variables.

The selected human position and orientation are transformed from rig to workspace coordinates, composed with the configured local tool correspondence, and then expressed in the robot base frame. These configuration offsets should not be described as a measured anatomical or hand–eye calibration. The cube-aware mode replaces the corresponding targets before the same robot optimisation.

3.5.2 Pose error and temporal terms

For target position $\mathbf{p}_t^*$ and orientation $\mathbf{R}_t^*$, the implemented task residual is

$$\mathbf{e}_t(\mathbf{q}_t) = \begin{bmatrix} \sqrt{w_p}\,[\mathbf{p}_m(\mathbf{q}_t, g_t) - \mathbf{p}_t^*] \\ \sqrt{w_R}\,\mathrm{Log}_{\mathrm{SO}(3)}\big((\mathbf{R}_t^*)^\top \mathbf{R}_m(\mathbf{q}_t, g_t)\big) \end{bmatrix}. \tag{3.9}$$

The rotation logarithm is represented as a three-vector. This is a weighted stack of translation and rotation errors, rather than the full $\mathbf{SE}(3)$ logarithm. The weights set the relative scale of metres and radians; they are not an estimated measurement covariance.

For $\Delta t = 1/f$, temporal operators use

$$(D_1\mathbf{q})_t = \frac{\mathbf{q}_{t+1} - \mathbf{q}_t}{\Delta t}, \qquad (D_2\mathbf{q})_t = \frac{\mathbf{q}_{t+2} - 2\mathbf{q}_{t+1} + \mathbf{q}_t}{\Delta t^2}. \tag{3.10}$$

The solver estimates f from median positive timestamp intervals and uses a uniform time step. Recordings with substantial irregular timing therefore require explicit resampling or a separate timing analysis. Velocity and acceleration penalties couple adjacent frames, and a posture penalty favours the selected robot reference. Temporal regularisation and interaction refinement are also examined in related work such as DexFlow [21]. Here, the discrete penalties

encourage smoothness; they do not establish continuous-time C^2 regularity, bounded jerk, or dynamic feasibility.

3.5.3 Confidence and robust data fitting

Let $c_{i,t} \in [0,1]$ be the prepared landmark confidence and let $\mathscr{P}$ contain wrist, index MCP, middle MCP, and pinky MCP. The default mean-palm evidence rule is

$$\bar{\omega}_t = v_t \left(\frac{1}{|\mathscr{P}|} \sum_{i \in \mathscr{P}} c_{i,t} \right)^{\alpha}, \tag{3.11}$$

where v_t is pose validity. Under the default episode-normalisation policy, triangulation quality is divided by the maximum quality in the episode before clipping. This relative normalisation should not be interpreted as absolute confidence across recordings. A stricter named profile uses

$$\bar{\omega}_t^{\text{all}} = \bar{\omega}_t \prod_{i \in \mathscr{P}} o_{i,t}, \tag{3.12}$$

where $o_{i,t}$ indicates direct observation. This all-observed rule is an alternative, not the inspected default. Under the mean rule a partly interpolated palm can retain positive pose weight.

Pinch-centre targeting further limits evidence by the lower fingertip

confidence. Invalid target frames receive zero data weight. The solver clips remaining weights to $[0,1]$ and applies a small positive floor to nonzero weights; the inspected default floor is 0.05. Consequently, arbitrarily weak positive evidence is not attenuated without limit. Position interpolation and rotation Slerp provide finite placeholders for invalid target rows, but those rows keep zero data weight.

A Huber loss is applied to the norm of the scaled six-dimensional residual. Its iteratively reweighted least-squares (IRLS) multiplier satisfies $\gamma_t^2 = \mathbf{min}(1, \delta / \mathbf{max}(\|\mathbf{e}_t\|$ Robust fitting reduces the influence of large residuals; it does not replace the interpolation and filtering already applied during preparation, and it cannot detect every plausible but incorrect hand pose.

3.5.4 Sparse batch solve and geometric constraints

The current solver optimises the whole arm trajectory using a custom sparse Gauss–Newton/IRLS procedure. With current stacked joint vector $\mathbf{q}$, pose Jacobians J_t, stacked reference $\mathbf{q}_{\text{ref}}$, and increment $\Delta\mathbf{q}$, its QP subproblem

minimises

$$\frac{1}{2}\Bigg[\sum_t \omega_t \gamma_t^2 \|\mathbf{e}_t + J_t \Delta\mathbf{q}_t\|^2 + \lambda_v \|D_1(\mathbf{q}+\Delta\mathbf{q})\|^2 + \lambda_a \|D_2(\mathbf{q}+\Delta\mathbf{q})\|^2 + \lambda_r \|\mathbf{q}+\Delta\mathbf{q}-\mathbf{q}_{\text{ref}}\|^2 + \mu\|\Delta\mathbf{q}\|^2\Bigg]. \tag{3.13}$$

This equation states the implemented local least-squares scaling. The code's separate logged merit uses the Huber data term plus the full weighted temporal and posture penalties; it is not numerically identical to the quadratic model in (3.13). The fixed damping μ is a Tikhonov term, rather than an adaptive Levenberg–Marquardt schedule.

Joint positions and velocities are constrained componentwise:

$$\mathbf{q}_{\min} \le \mathbf{q}_t \le \mathbf{q}_{\max}, \tag{3.14}$$

$$-\dot{\mathbf{q}}_{\max} \le (D_1\mathbf{q})_t \le \dot{\mathbf{q}}_{\max}. \tag{3.15}$$

The geometric constraints have the form

$$h_j(\mathbf{q}_t, g_t) \ge 0, \tag{3.16}$$

with local linearisation inside each QP. They cover the workspace floor plane

for represented moving bodies, including the adapter and gripper contact points, and enabled self-collision pairs. The calibrated plane is transformed into robot coordinates, including for a tilted base. These checks do not include arbitrary scene obstacles, general robot–object contact constraints, torque limits, or hard acceleration and jerk limits.

Pinocchio supplies kinematics and Jacobians, Pink supplies the configured self-collision barriers [25], and qpsolvers invokes the sparse OSQP backend. Pink does not itself formulate the entire batch objective. The trajectory starts from the tiled reference posture, normally the registered robot home, rather than a sequential frame-to-frame IK warm start. Each QP also limits increment magnitude. Backtracking checks the nonlinear merit and represented geometric margins once the current trajectory is geometrically feasible; an initially infeasible trajectory may require a feasibility-repair step.

The final trajectory is checked again against exact represented floor and self-collision margins. The inspected floor tolerance is $0.5\,\mathrm{mm}$; it is a numerical acceptance tolerance for the model, not a physical clearance guarantee. A small accepted increment can trigger the solver's convergence flag even if tracking error remains large. Convergence, constraint satisfaction, target-tracking error, and task success are therefore separate outcomes.

3.5.5 Approach segment and matched baseline

After retargeting, a quintic approach connects the registered home posture to the first demonstration configuration. Its duration is increased as needed to satisfy velocity limits, and the concatenated trajectory is checked for represented joint, velocity, floor, and enabled collision constraints. The approach is generated robot motion rather than a measured part of the human episode. It must be separated from demonstration frames when aligning video, object motion, or accuracy statistics.

An optional comparison path runs causal framewise IK followed by Savitzky–Golay smoothing. A meaningful comparison uses the same targets, robot assembly, gripper sequence, and data weights, and reports resulting constraint violations as well as tracking error. Smoothing a sequential solution does not by itself preserve its feasibility.

3.6 Proposed 3DGS Reconstruction and Robot Rendering

3.6.1 Scene decomposition and asset acquisition

The proposed appearance branch uses 3DGS for reusable scene appearance [5], combined with separately represented moving objects and a robot model.

This hybrid design follows the general use of reconstructed assets in human-to-robot data generation [2], [7]; it is not presented as a previously absent combination of components.

The initial target is a static workspace background, a small set of rigid task objects, and one articulated robot. Human masks exclude the demonstrator from background fitting. An empty-scene scan or additional views are needed where the human or task objects otherwise hide the background. Moving objects are reconstructed in canonical coordinates and positioned using their estimated tracks. A multiview point cloud or a set of segmentation masks is an input to this process, not already a trained Gaussian scene.

For the first implementation, the background is to use 3DGS and the robot is to use its articulated visual meshes. Task objects may use scanned meshes or rigid Gaussian assets, with the representation fixed for each comparison. All assets share the calibrated metric frame. Additional scans, segmentation corrections, and asset preparation contribute to the data-collection cost even though they do not require per-robot teleoperation.

3.6.2 Reconstruction objective and supported views

A candidate reconstruction objective for scene parameters Θ is

$$\begin{aligned}\mathscr{L}(\Theta) =& \lambda_{\text{rgb}}\mathscr{L}_{\text{rgb}}(M \odot \hat{I}, M \odot I) + \lambda_{\text{ssim}}\mathscr{L}_{\text{ssim}}(\hat{I}, I; M) \\ &+ \lambda_z \frac{\sum_u M_z(u)\,|\hat{z}(u) - z(u)|}{\sum_u M_z(u)}.\end{aligned} \tag{3.17}$$

Here M selects valid observations of the modelled component, while M_z additionally excludes invalid depth and depth-boundary regions. The depth term is evaluated only when valid pixels exist. A consistent camera-axis depth convention is required for both measured and rendered depth. The masked structural term must avoid windows crossing excluded regions. Loss weights, Gaussian initialisation, optimisation schedule, and any depth regularisation remain experimental choices to be specified with the implementation; no performance is assumed from this formulation.

Novel cameras are sampled within a declared operating region supported by the captured geometry. The method does not assume that a small fixed rig observes every surface. Disoccluded or poorly supported regions are marked and included in rejection or coverage statistics. Hallucinated completion, if later introduced, would require a separately identified method and evaluation.

3.6.3 Time-consistent robot compositing

At demonstration time t, the renderer receives the retargeted arm state, the profiled gripper command, object poses, camera intrinsics and extrinsics, and appearance parameters. Robot link transforms are computed from the same model and joint ordering used for the exported state. Rendering uses the achieved forward-kinematics pose, rather than drawing the end-effector at an unreachable human target.

The compositor must resolve visibility between robot meshes, moving objects, and the Gaussian background in a consistent depth convention. A foreground overlay without scene-depth tests is insufficient. The chosen treatment of Gaussian opacity and depth must be specified and checked at contact boundaries and thin structures. RGB, depth, component masks, and camera metadata should be generated together when supported.

Object motion initially remains a prescribed estimate from the source episode. Displaying an object next to the gripper does not establish that the gripper can cause that motion. Penetration and contact-region checks can reject geometric inconsistencies, but physical manipulation success would require a contact simulation or hardware experiment outside this study's evaluation scope.

3.7 Factorial Generation and Episode Acceptance

For source episode d, let $m \in \{1,\dots,M\}$ denote embodiment, $k \in \{1,\dots,K\}$ a camera configuration, and $p \in \{1,\dots,P\}$ an appearance configuration. The proposed output is a sequence

$$\mathscr{E}_d^{m,k,p} = \left\{ \hat{I}_{d,t}^{m,k,p}, \mathbf{q}_{d,t}^{m}, g_{d,t}^{m}, \mathbf{T}_{O,d,t}^{C}, \tau_{d,t}, \mathscr{V}_{d,t}^{m,k,p} \right\}_{t=1}^{T_d}, \tag{3.18}$$

where $\mathscr{V}$ stores relevant validity and support information. Embodiment-specific retargeting is performed before varying cameras and appearance, so the same accepted robot trajectory can support multiple visual observations. Appearance variation is initially restricted to declared edits that preserve geometry and visibility checks. A lighting change is not automatically physically valid for a Gaussian model with baked appearance. Any edit that changes task geometry requires new retargeting and interaction checks.

The Cartesian product gives MKP requested configurations, not a guaranteed number of usable demonstrations. Define $A_d^{m,k,p} \in \{0,1\}$ for configurations that pass the stated checks; then

$$N_{d,\text{accepted}} = \sum_{m=1}^{M} \sum_{k=1}^{K} \sum_{p=1}^{P} A_d^{m,k,p} \leq MKP. \tag{3.19}$$

Acceptance policies should be fixed before the final comparison and retain failure reasons. They must distinguish insufficient hand/object evidence, robot constraints or unacceptable tracking error, unsupported camera views, visible geometric interaction defects, and inconsistent data alignment. Thresholds are experimental parameters, not values inferred from the existence of a processing stage. Current trajectory-export eligibility is weaker than this proposed synthetic-episode acceptance.

All variants of a source episode share its original motion evidence and are statistically dependent. Dataset splits must therefore be made by source episode, and where appropriate by subject, object, or scene, before factorial expansion. Viewpoint or appearance changes increase observation variation; they do not create independent manipulation strategies. Counts, yield, processing time, and manual preparation effort are reported per source and per factor combination.

3.8 Evaluation Protocol

3.8.1 External hand and object references

HO-Cap supplies an external multiview RGB-D reference for perception [10]. The protocol specified for the selected 28-sequence single-hand pick-and-place cohort uses all eight RealSense views and excludes the HoloLens stream. This cohort is a task/handedness subset, not an official held-out split or the full dataset. Reproducing its results requires the exact sequence list, dataset revision, frame counts, profile manifest, and completed-stage records. The inspected application source alone does not establish that a benchmark run was completed.

Reference hand labels are excluded from detection, reconstruction, and parameter fitting. Predictions and references are compared in a common metric frame by exact frame identity, without root alignment, Procrustes fitting, temporal search, or reference-based sample selection. The hand measures are MPJPE, PCK@20/50 mm, P95 error, and prediction coverage. Directly observed, interpolated, and articulated-fit outputs are distinguished. Matched-joint-weighted aggregate means and PCK values are reported alongside per-sequence statistics; averaging sequence quantiles does not produce a pooled

P95. HO-Cap's semi-automatic MANO-based annotations are reference labels, not independent motion-capture ground truth or a competing estimator run inside ViKi.

The reference-assisted object protocol supplies published instance masks and depth to the tracker. It therefore tests tracking conditional on segmentation, separately from the autonomous SAM-based branch. One constant object-local transform is fitted at the first confident frame using the reference pose; subsequent translation and geodesic rotation errors measure relative tracking drift under that initial alignment. The documented confidence threshold is 0.35. Reference-assisted initial alignment does not measure absolute pose localisation. Coverage, stationary/moving subsets, and object symmetries must be stated; a raw geodesic score without symmetry handling has that limitation.

Benchmark error and local-rig self-consistency remain separate evidence. Neither a small calibration residual nor agreement with a fitted hand model substitutes for comparison with an independent reference.

3.8.2 Retargeting and interaction comparisons

For each actually tested robot assembly, the evaluation records base placement, model revision, tool correspondence, solver settings, and source profile. It measures TCP position and rotation error, accepted frames and episodes, joint and velocity-limit violations, and represented floor/self-collision margins. Error is reported separately on evidence-supported and filled targets. Solver convergence flags are reported alongside these measures, rather than used as their substitute.

Hand-only and cube-aware retargeting use matched source episodes and robot assemblies. Object-relative TCP error, contact-region alignment, closure timing, and geometric penetration characterise their interaction consistency. Gripper commands are compared against the declared decoder and actual modelled pad gap. These measures do not become grasp-success rates without an executed interaction. General object-relative transfer requires its own implementation and matched comparison before a benefit can be claimed.

Dense source trajectories preserve more timing information than sparse keyframes, but filters, gap filling, gripper decoding, reference-posture penalties, and cube-specific orientation replacement can change the motion. Where natural-motion preservation is investigated, it is measured through timing,

path, and orientation differences on supported intervals; it is not assumed from markerless capture.

3.8.3 Rendering and expansion experiments

The proposed reconstruction evaluation holds out cameras and scene states before tuning. PSNR, SSIM, and valid-pixel depth RMSE compare corresponding scene components in matching states, with background and task-object regions reported separately. A static background model is not evaluated against unmasked moving humans. A human demonstration image is also not a paired ground-truth image of the replacement robot.

Compositing comparisons use identical trajectories, object states, and camera poses. They inspect depth ordering, robot/object boundaries, temporal stability, and agreement between rendered geometry and exported state. Paired perceptual metrics require a corresponding target image; otherwise the evaluation uses explicitly defined geometric and consistency tests rather than a misleading full-image similarity score.

Factorial expansion is compared with single-configuration generation on the same source episodes. The report includes requested and accepted combinations, rejection categories, supported viewpoint range, runtime, and

manual intervention. Any comparison with teleoperation effort must include capture, calibration, scans, corrections, robot-model preparation, computation, and rejected outputs. Policy-training benefit is not inferred from appearance diversity or episode count.

3.9 Export, Provenance, and Replay Status

The working default exporter writes a self-contained NumPy trajectory bundle and JSON metadata. It retains joint trajectories and derivatives, normalised gripper opening, modelled metric opening, target and achieved poses, tracking errors, source timing, and evidence information. Optional object-model and post-IK object-relative artifacts are copied when their stages are complete. Units, frame names, source episode, processing profile, robot assembly, and parameter provenance must remain available. In the inspected plan schema, the metric gripper opening is based on the URDF pad-centre gap; older schemas may use nominal width and must be interpreted by version.

Current export requires a readable plan, not a demonstrated manipulation outcome. It may include unlabelled episodes and reports missing or skipped

artifacts in its manifest. The optional LeRobot path has stricter eligibility and partial annotation support; shape-correct placeholders are not measured object-relative labels. The existing bundle is therefore a trajectory artifact, rather than the completed rendered dataset in (3.18).

For the proposed synthetic export, every rendered frame must be paired with the state that produced it. The schema must distinguish state at t from a command for $t+1$ and specify whether an action is an absolute joint target, an increment, or a velocity command. Camera calibration, appearance seed, renderer and asset versions, validity masks, and acceptance decisions are additional provenance. Original human-camera videos cannot be relabelled as robot observations merely because a robot trajectory has been generated.

3.9.1 Physical replay

The current dry-run replay path echoes a plan and does not measure robot tracking. The physical driver and several screening checks remain unimplemented. No physical-replay acceptance or achieved hardware state can therefore be inferred from a dry-run result. Hardware execution, contact simulation, and policy training lie outside the present reconstruction, retargeting, and rendering evaluation. They would supply additional evidence needed for claims about

executable manipulation or learning effectiveness.

3.10 Assumptions and Operating Limits

The initial scope is a fixed, calibrated RGB-D rig, one tracked hand, rigid task objects, a bounded workspace, and parallel-jaw transfer. Handedness, object identities, task roles, and robot placement must be declared. Unknown geometry and uncertain observations remain explicit rather than being treated as valid because an intermediate file exists.

The reconstruction assumes sufficient visibility and stable calibration. The retargeter assumes that the selected human-to-tool correspondence and robot model are appropriate for the task. Its collision checks cover only the represented geometry. The proposed appearance branch assumes a separable static background and rigid moving assets; deformable objects, general scene relighting, and arbitrary unseen surfaces require further models and validation.

The intended outcome is a measured set of geometrically consistent candidate robot episodes derived without per-robot teleoperation during human capture. Whether a source episode supports multiple embodiments, views, and scene appearances is an experimental result. Preservation of task-relevant

human motion, physical manipulability, and downstream learning benefit are separate claims with separate evidence requirements.

Глава 4

Implementation and Evaluation

4.1 Software and Experimental Setup

The Python implementation separates capture, calibration, perception, preparation, retargeting and export. FastAPI serves offline jobs and the browser inspection interface; MuJoCo tools reside in a separate image. Each stage writes a durable episode artifact. The rec.npz hand extraction, cln.npz prepared motion, object_models.npz scene model and plan.h5 robot plan represent different stages and are not interchangeable evidence. Physical-robot replay is a stub and is not evaluated.

The local apparatus has two Azure Kinect DK cameras on independent

USB controllers. The intended tabletop task is to transfer a blue cube from a red support to a green target, using a simulated old-generation UR10 with a Robotiq 2F-85 gripper. No physical UR10 is available for this study. The earlier local cube recordings and their derived artifacts are not available for verification, so no numerical result from that campaign is included in this manuscript. In particular, local capture-synchronisation, stage-yield, object-model, IK, contact-replay and learned-policy results are not claimed. A new, preserved local cohort would be required to evaluate that end-to-end chain.

4.2 External HO-Cap Hand and Object Evaluation

The full external experiment covers 28 single-hand HO-Cap pick-and-place sequences from nine subjects: 24 right-hand and four left-hand, with 29,706 frames from eight calibrated RealSense views. These are not the local Kinect cube recordings. The source revision and all 114 numerical JSON files are preserved in the paper's pre-retarget results snapshot with SHA-256 hashes. Capture and object protocols are documented in the accompanying dated logs. Both processing queues completed 28/28 scenes. This is a selected task cohort, not an evaluation on the official HO-Cap held-out split [10].

4.2.1 Hand-Joint Accuracy

Every one of the 623,826 valid reference joint slots has a finite prepared prediction. Matched-joint-weighted MPJPE is 10.79 mm and PCK@20/50 mm is 92.77/99.85%. The per-sequence MPJPE range is 9.26–12.85 mm; the per-sequence P95 range is 17.72–31.90 mm. No pooled P95 is inferred from these 28 quantiles. The four left-hand scenes have weighted MPJPE 11.60 mm, versus 10.65 mm for 24 right-hand scenes; handedness and subject are confounded here, so this is descriptive rather than a controlled difference. The fused-observed stage, before final temporal smoothing, matches 623,799/623,826 slots (99.996%) with weighted MPJPE 11.16 mm. The all-finite smoothed score includes 27 filled slots and should not be mistaken for direct observation coverage. Table 4.1 summarizes the cohort; all 28 episode rows, including P95 and PCK, appear in Table B.1.

TABLE 4.1
Eight-view HO-Cap 21-joint accuracy against published MANO fits

Subset and stage	Scenes	Matched / GT	MPJPE (mm)	PCK@20 (%)
All, fused observed	28	623,799/623,826	11.16	91.68
All, smoothed all	28	623,826/623,826	10.79	92.77
Left, smoothed all	4	86,667/86,667	11.60	87.56
Right, smoothed all	24	537,159/537,159	10.65	93.61

4.2.2 Named Object Tracks and Outliers

Each scene contains four annotated objects, giving 112 object–sequence tracks across 15 object groups. The official HO-Cap object descriptors hocap-catalog are keyed by IDs in Table B.3; for example, G18_2 is a projector remote, not an anonymous object. All four instances in the G18 scenes are the green tea latte mix, projector remote, right shoe and left shoe. This identity mapping comes from the authors' hocap_info.yaml, rather than appearance-based naming.

The tracker covers 117,916 of 118,824 valid reference object frames (99.24%); 97/112 tracks have complete coverage. Across the 112 track means, the median and arithmetic mean relative translation errors are 4.74 and 5.44 mm; the corresponding rotation values are 3.47 and 7.58 degrees. Frame-count-weighted averages of the track means are 5.60 mm and 7.55 degrees. These are not pooled frame-level P95 values. The median of the 112 track P95s is 10.60 mm translation and 9.25 degrees rotation. First-frame alignment is fitted using the reference pose, so the numbers measure subsequent relative drift rather than absolute object-frame accuracy. The supplied instance masks remove segmentation errors from this test; static intervals remain mixed with manipulated intervals.

TABLE 4.2
Selected HO-Cap object failures; mean and P95 are within-track errors

Sequence	Object and official descriptor	Cov. (%)	T mean	T P95	R mean	R P95
S7/162803	G18_2 projector remote	98.63	12.38	20.44	112.60	175.26
S5/112303	G15_2 dandelion tea	100.00	17.24	21.81	109.65	128.43
S2/202617	G09_3 toothpaste box	100.00	5.77	13.50	74.03	132.88
S7/193506	G02_4 sauce bottle	99.65	2.82	10.67	56.57	169.89
S2/163929	G19_1 electric screwdriver	48.80	5.75	15.41	10.55	50.95
S2/201942	G06_1 DVD	100.00	19.39	25.24	1.90	5.63

The worst rotation case is the projector remote in subject_7/162803: mean 112.60 degrees and P95 175.26 degrees despite 98.63% tracked coverage and 12.38 mm mean translation error. In another G18 scene (subject_1/165502), the same object ID has 6.97-degree mean rotation error, and the other three objects in the failing scene have 1.10–1.34-degree means. Thus the anomaly is sequence/object-specific rather than a global frame rotation. A visual review showed plausible late object placement under an alternate orientation, but did not establish a unique root cause; the outlier remains in all aggregates. The dandelion tea (109.65 degrees) and toothpaste box (74.03 degrees in one scene, versus 13.82 and 6.89 degrees in two others) show that this is not a single isolated failure. A low translation error also does not certify orientation, as the sauce bottle demonstrates. Conversely, the DVD has the highest mean translation error, 19.39 mm, with only 1.90-degree mean rotation error. The electric screwdriver is a coverage failure at 48.80%, so its conditional error on visible frames must not stand in for whole-sequence accuracy. Tables B.2

and B.3 retain all 112 object scores and official identities for audit.

4.2.3 Historical Two-Sequence Camera-Pair Pilot

Before the full eight-view run, two subject-1 sequences were tested for 120 frames each with a pair selected after screening all 28 pairs on those same frames. Table 4.3 is retained as a historical in-sample development comparison; its original local artifacts were lost. The full-sequence eight-view results above supersede it as the reproducible hand/object reference cohort, but do not turn the selected pair into a held-out baseline.

TABLE 4.3
Historical HO-Cap Hand-Joint Pilot: Selected Pair Versus Eight Views

Sequence	Views	Coverage	MPJPE (mm)	P95 (mm)	PCK@20 (%)
170650	selected pair	2,520/2,520	11.33	22.71	87.70
170650	all eight	2,520/2,520	11.34	22.17	87.50
170105	selected pair	2,520/2,520	11.67	22.80	89.80
170105	all eight	2,520/2,520	10.50	20.15	94.88

The selected pair also failed one colour–depth consistency check. No HO-Cap score establishes the accuracy of the local Kinect calibration or physical success of the proposed robot motion.

4.3 Downstream Evidence Boundary

The archived HO-Cap experiment evaluates hand reconstruction and first-frame-aligned object tracking on a separate RealSense capture system. It does not establish the accuracy of the local Kinect rig or produce a validated robot demonstration. Retarget plans, simulator contacts, task success and policy learning require their own retained inputs, protocols and numerical artifacts before they can be reported as results. This chapter therefore ends at the external perception evaluation rather than extrapolating an end-to-end success rate.

Глава 5

Analysis and Discussion

5.1 What the Results Establish

The retained eight-view HO-Cap evaluation covers 28 selected pick-and-place sequences from nine subjects. Prepared 21-joint hand predictions match all 623,826 valid reference joint slots, with 10.79 mm matched-joint-weighted MPJPE against the dataset's published semi-automatic MANO fits. This is an external-system perception measurement, not evidence that the local Azure Kinect rig has the same accuracy. The object evaluation covers 112 named tracks, but it uses published instance masks and a reference-assisted first-frame SE(3) alignment. Its 5.44 mm translation and 7.58-degree rotation

macro-means describe relative tracking drift conditional on those inputs, not absolute object localization or autonomous segmentation.

5.2 Error Sources and Reference Frames

The hand and object numbers cannot be added into an end-to-end geometric error budget. The hand score is an absolute comparison in the HO-Cap world frame; the object score removes a per-object initial gauge difference. Both refer to semi-automatic annotations rather than an independent metrology system. The object tracker also receives masks generated outside ViKi, and static intervals are included alongside manipulation. Thus a small average object drift need not imply a sufficiently accurate grasp pose during contact.

The outliers make this distinction operational. The projector remote G18_2 has 112.60-degree mean rotation error in one sequence at 98.63% coverage, but only 6.97 degrees in another. The dandelion tea and toothpaste box have additional large orientation errors. The sauce bottle combines 2.82 mm mean translation error with 56.57-degree mean orientation error. These tracks remain in the aggregate rather than being removed as failures. Symmetry-

aware orientation scoring was not used; the visual review of the remote did not uniquely diagnose the failure. Conversely, the electric screwdriver is tracked for only 48.80% of valid reference frames, so its conditional error does not describe the missing half.

For the local two-Kinect system, camera geometry, RGB–depth alignment, exposure timing and table-height registration remain unquantified in this manuscript. The former local numerical summaries cannot be rechecked because their source recordings and derived artifacts are unavailable. Code paths and visually plausible clouds are not substitutes for a preserved evaluation protocol.

5.3 Statistical Validity and Attribution

The HO-Cap selection is a single-hand pick-and-place subset, not the official held-out benchmark split. Subjects and hands are unevenly represented: 24 right-hand and four left-hand sequences. Joint slots within the same frames and object frames within the same sequence are correlated, so their large counts do not imply that many independent trials. The two-sequence camera-pair pilot was selected on the same development frames and cannot be

interpreted as held-out evidence. The main eight-view aggregate is descriptive for this selected cohort; no causal claim about the benefit of camera count, training data or robot embodiment follows from it.

No surviving local cube dataset supports the previous downstream figures. In particular, there is no verifiable local stage yield, simulated stacking rate, policy comparison or contact-safety result. The current manuscript therefore does not claim an end-to-end camera-to-simulation proof of concept or Real-to-Sim-to-Real transfer.

5.4 Scope and Implications

The supported contribution is a reproducible external perception assessment, with explicit object identities, coverage, outliers and metric limitations, alongside an implemented offline software pipeline. The intended cube-transfer experiment remains a research question rather than a completed result. A new preserved local cohort should include measurement-only controls: known geometry at table and carry heights, a marked asymmetric object for orientation, and disjoint human trials. Retained simulator rollouts and matched no-demonstration baselines would then be needed before reporting physical task success or

learning benefit. No physical UR10 evaluation is part of the current evidence.

Глава 6

Conclusion

6.1 Summary

ViKi implements an offline route from multi-view RGB-D capture to separate hand, object and candidate robot-plan artifacts. This thesis reports the retained, reproducible HO-Cap perception evaluation, not an end-to-end robot-learning result. The earlier local cube recordings and their downstream artifacts are unavailable, so their numerical summaries and derived simulation or policy claims are excluded.

6.2 Answers to the Research Questions

RQ1: pipeline yield. The external eight-view HO-Cap evaluation completed on 28 selected sequences and 29,706 frames. Its prepared hand tracks have 10.79 mm matched-joint-weighted MPJPE against the published semi-automatic reference, with 623,826/623,826 valid joints matched. For 112 named object tracks, macro-mean relative translation and rotation drift are 5.44 mm and 7.58 degrees after first-frame alignment and with published instance masks. Large orientation failures remain, including 112.60 degrees mean error on one projector-remote track. These data quantify external perception, not the local Kinect-to-robot pipeline yield or absolute object pose accuracy.

RQ2: object-aware transfer. No retained local demonstration and matched simulator evaluation supports a claim about physics-validated transfer to new cube placements. The question remains open.

RQ3: learning benefit. No reproducible comparison against a matched no-ViKi learner is available. The benefit of ViKi-derived examples or their sample efficiency is not established.

RQ4: visual policy. No evaluated image-conditioned robot policy is reported.

6.3 Claim Boundary and Next Validation

The current evidence supports a measurable HO-Cap hand-reconstruction result and a conditional, relative object-tracking result. It does not validate local camera calibration, complete cube-transfer demonstrations, collision-free grasping, physical simulator success, policy generalization or real-robot transfer. Re-recording a preserved local cohort with explicit calibration controls is necessary before testing those claims. Each subsequent retarget, physics and learning result must retain its input scenes, protocol and numerical outputs so it can be independently checked.

6.4 Concluding Remarks

The distinction between a reconstructed hand, a tracked object, a geometric robot plan and a successful manipulation is essential. Only the first two have quantitative external evidence in this manuscript; the latter stages remain work to evaluate.

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Приложение A

Notation

TABLE A.1
Symbols Used Throughout the Thesis

Symbol	Meaning
W	workspace frame, defined by the ChArUco anchor
C_k	frame of camera k
R	manipulator base frame
E	manipulator end-effector frame
H	human wrist frame
O	manipulated object frame
$\mathbf{T}_B^A$	rigid transform from frame B to frame A, element of $\mathbf{SE}(3)$
$\mathbf{q}_t \in \mathbb{R}^n$	manipulator configuration at timestep t
$f(\mathbf{q}_t)$	forward kinematics of the end-effector
$\mathbf{e}_t$	task-space residual, (3.9)
ω_t	per-frame perception confidence weight, (3.11)
$q_{i,t}$	geometric quality of a triangulated landmark, (3.4)
$w_{i,t}^k$	per-view weight, single-view fallback, (3.5)
$v_{i,t}^k$	detector visibility score
$c_{i,t}^k$	sensor depth confidence
ρ_δ	Huber loss with transition parameter δ
$\lambda_v, \lambda_a, \lambda_r$	velocity, acceleration and posture weights
g_t	continuous gripper opening, (3.6)
τ	host-side monotonic timestamp, (3.2)
T	number of timesteps in an episode

Приложение B

Full HO-Cap Perception Results

Sequence labels use S<subject>/<HHMMSS> from the original HO-Cap identifier. The complete unrounded JSON and checksums are preserved in paper/results/hocap/pre_retarget/. Tables in this appendix can be regenerated with python3 scripts/build_hocap_paper_tables.py; the script refuses modified source hashes or an incomplete cohort. Object descriptors follow the official HO-Cap toolkit's config/hocap_info.yaml hocap-catalog. Hand P95 is per sequence; object P95 is per object–sequence track, never pooled across rows.

TABLE B.1
Eight-view HO-Cap hand results by sequence; PCK values are percentages.

Sequence	Frames	Matched	MPJPE	P95	PCK@20	PCK@50
S1/165502	676	14196	12.85	25.48	82.83	99.87
S1/165807	798	16758	12.10	24.18	84.35	99.79
S1/170105	831	17451	11.61	21.46	92.71	99.91
S1/170650	741	15561	12.55	23.78	86.36	100.00
S1/170959	874	18354	11.26	21.67	92.45	99.96
S2/201316	828	17388	10.77	21.23	93.48	99.95
S2/201942	1561	32781	11.86	23.79	86.87	99.95
S2/202617	1271	26691	11.06	22.30	91.43	99.99
S2/163929	1129	23709	10.48	20.43	94.11	99.99
S3/154531	1686	35406	9.26	18.09	96.95	99.98
S3/155008	1046	21966	9.45	17.72	97.89	100.00
S3/161209	1071	22491	9.71	18.75	96.10	99.92
S3/161937	1269	26649	10.33	19.48	95.74	99.79
S3/162327	878	18438	9.82	19.30	95.79	99.98
S3/162756	946	19866	9.56	17.79	98.23	100.00
S4/163223	939	19719	12.54	30.34	86.51	99.24

Sequence	Frames	Matched	MPJPE	P95	PCK@20	PCK@50
S4/164812	974	20454	10.57	20.61	94.22	99.99
S5/112303	702	14742	10.95	26.74	91.05	99.54
S6/110646	1292	27132	10.02	19.26	95.51	99.98
S6/111118	1272	26712	10.54	21.84	93.52	99.87
S6/112546	1311	27531	10.06	21.66	93.83	99.69
S7/193506	1136	23856	12.40	31.90	87.20	98.66
S7/162803	1092	22932	10.10	21.11	93.81	99.86
S8/180651	985	20685	11.88	24.84	89.59	99.88
S9/123403	992	20832	10.28	19.54	95.56	99.99
S9/123725	1134	23814	10.95	20.10	94.81	99.99
S9/124057	1136	23856	10.88	21.39	93.45	99.84
S9/125315	1136	23856	10.74	20.45	94.53	99.97

TABLE B.2
Per-object relative-pose tracking after one first-frame SE(3) alignment. T and R are translation in mm and geodesic rotation in degrees.

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S1/165502	G18_1	100.00	2.68	3.75	1.32	1.97

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S1/165502	G18_2	100.00	7.70	24.15	6.97	27.41
S1/165502	G18_3	100.00	1.49	6.97	1.32	7.71
S1/165502	G18_4	100.00	1.66	7.76	2.20	5.86
S1/165807	G15_1	100.00	2.41	10.53	1.58	3.85
S1/165807	G15_2	100.00	2.00	2.76	3.02	3.34
S1/165807	G15_3	100.00	2.17	10.08	1.24	7.77
S1/165807	G15_4	100.00	6.02	12.56	10.65	18.60
S1/170105	G10_1	100.00	0.67	3.56	1.55	10.44
S1/170105	G10_2	100.00	5.57	6.37	11.96	20.04
S1/170105	G10_3	100.00	10.13	23.41	3.54	9.66
S1/170105	G10_4	100.00	1.23	4.46	1.06	3.36
S1/170650	G04_1	100.00	1.41	4.74	1.64	3.84
S1/170650	G04_2	100.00	2.21	3.05	1.08	2.63
S1/170650	G04_3	100.00	8.89	20.10	6.02	14.70
S1/170650	G04_4	100.00	0.72	2.98	0.32	2.15
S1/170959	G19_1	100.00	2.41	8.12	4.74	15.42
S1/170959	G19_2	100.00	12.85	46.05	3.41	8.96

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S1/170959	G19_3	100.00	2.95	10.80	5.50	19.73
S1/170959	G19_4	100.00	14.36	28.21	13.02	65.53
S2/201316	G05_1	100.00	4.73	12.06	3.30	9.27
S2/201316	G05_2	100.00	0.94	4.74	0.74	4.64
S2/201316	G05_3	100.00	9.40	28.11	7.81	17.53
S2/201316	G05_4	100.00	6.03	12.08	4.18	12.11
S2/201942	G06_1	100.00	19.39	25.24	1.90	5.63
S2/201942	G06_2	100.00	0.55	4.18	0.67	7.04
S2/201942	G06_3	100.00	6.42	10.68	7.26	17.51
S2/201942	G06_4	99.68	7.40	28.59	13.03	40.35
S2/202617	G09_1	100.00	11.27	24.44	3.91	18.56
S2/202617	G09_2	100.00	3.93	8.17	3.14	6.62
S2/202617	G09_3	100.00	5.77	13.50	74.03	132.88
S2/202617	G09_4	99.84	13.28	31.71	5.18	12.13
S2/163929	G19_1	48.80	5.75	15.41	10.55	50.95
S2/163929	G19_2	100.00	9.67	25.64	1.94	5.60
S2/163929	G19_3	100.00	2.13	7.22	6.38	8.89

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S2/163929	G19_4	100.00	10.74	34.86	2.20	6.96
S3/154531	G09_1	100.00	5.93	13.05	2.77	9.75
S3/154531	G09_2	100.00	6.10	13.35	6.79	28.73
S3/154531	G09_3	100.00	9.03	14.95	13.82	142.95
S3/154531	G09_4	100.00	7.41	17.99	4.77	7.02
S3/155008	G10_1	100.00	1.59	4.16	1.93	7.28
S3/155008	G10_2	100.00	5.55	12.66	3.94	17.99
S3/155008	G10_3	100.00	5.67	7.58	1.70	6.88
S3/155008	G10_4	100.00	4.51	8.15	2.31	9.23
S3/161209	G16_1	100.00	1.91	4.31	5.78	6.44
S3/161209	G16_2	100.00	4.74	11.53	13.93	90.87
S3/161209	G16_3	100.00	2.86	7.87	1.92	10.59
S3/161209	G16_4	100.00	9.37	20.60	9.55	41.50
S3/161937	G19_1	100.00	3.11	16.36	7.25	51.05
S3/161937	G19_2	100.00	5.98	14.12	1.52	5.72
S3/161937	G19_3	100.00	3.50	7.80	7.81	18.02
S3/161937	G19_4	100.00	18.47	45.93	4.65	12.46

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S3/162327	G20_1	100.00	0.99	2.36	0.93	2.31
S3/162327	G20_2	100.00	4.66	10.04	3.04	7.04
S3/162327	G20_3	100.00	1.93	9.06	0.62	2.82
S3/162327	G20_4	95.22	6.53	14.24	6.58	21.72
S3/162756	G21_1	100.00	2.51	4.55	2.18	3.26
S3/162756	G21_2	100.00	3.91	14.02	8.22	52.38
S3/162756	G21_3	100.00	10.18	22.65	2.89	6.76
S3/162756	G21_4	99.89	4.70	10.28	4.46	29.13
S4/163223	G16_1	100.00	5.17	23.66	5.12	25.23
S4/163223	G16_2	100.00	3.78	9.71	4.00	9.65
S4/163223	G16_3	100.00	3.58	5.05	10.68	15.81
S4/163223	G16_4	93.93	9.75	28.59	9.55	35.11
S4/164812	G21_1	100.00	1.72	3.03	1.23	3.54
S4/164812	G21_2	100.00	2.86	10.76	6.74	16.99
S4/164812	G21_3	100.00	15.06	32.43	3.60	10.91
S4/164812	G21_4	100.00	7.72	13.79	5.29	19.97
S5/112303	G15_1	100.00	1.12	5.40	1.91	13.57

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S5/112303	G15_2	100.00	17.24	21.81	109.65	128.43
S5/112303	G15_3	100.00	5.28	8.30	8.28	12.34
S5/112303	G15_4	100.00	4.63	11.75	8.16	27.42
S6/110646	G05_1	100.00	2.76	6.17	1.83	5.31
S6/110646	G05_2	100.00	1.77	3.99	0.69	1.72
S6/110646	G05_3	91.33	5.27	21.45	4.69	28.75
S6/110646	G05_4	98.22	7.22	17.21	9.47	24.79
S6/111118	G06_1	99.37	10.02	18.20	2.08	7.79
S6/111118	G06_2	100.00	1.49	2.64	5.78	8.36
S6/111118	G06_3	100.00	1.33	3.50	2.61	6.06
S6/111118	G06_4	96.31	8.01	13.28	41.23	65.31
S6/112546	G22_1	99.31	6.76	15.41	8.75	82.09
S6/112546	G22_2	100.00	1.77	4.50	1.49	2.46
S6/112546	G22_3	100.00	0.60	2.39	0.95	2.31
S6/112546	G22_4	100.00	10.98	30.93	2.42	7.98
S7/193506	G02_1	100.00	7.51	8.67	2.30	5.40
S7/193506	G02_2	99.65	4.85	7.52	4.64	11.53

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S7/193506	G02_3	100.00	1.10	5.58	1.12	6.37
S7/193506	G02_4	99.65	2.82	10.67	56.57	169.89
S7/162803	G18_1	100.00	5.49	7.02	1.33	4.87
S7/162803	G18_2	98.63	12.38	20.44	112.60	175.26
S7/162803	G18_3	100.00	0.98	6.86	1.10	6.94
S7/162803	G18_4	100.00	1.64	4.22	1.34	4.97
S8/180651	G06_1	100.00	15.36	37.92	2.05	6.07
S8/180651	G06_2	100.00	0.87	4.45	0.77	3.83
S8/180651	G06_3	100.00	3.04	7.80	2.23	5.96
S8/180651	G06_4	99.90	5.01	13.93	12.31	35.29
S9/123403	G07_1	100.00	5.00	9.78	4.49	18.84
S9/123403	G07_2	100.00	0.44	2.22	0.32	1.99
S9/123403	G07_3	100.00	1.05	3.60	2.45	8.26
S9/123403	G07_4	100.00	2.80	5.69	1.90	3.75
S9/123725	G09_1	100.00	11.53	26.72	3.78	14.00
S9/123725	G09_2	100.00	6.25	16.46	4.82	17.28
S9/123725	G09_3	100.00	5.51	13.62	6.89	36.18

Sequence	Object	Cov. (%)	T mean	T P95	R mean	R P95
S9/123725	G09_4	100.00	1.92	3.65	1.32	3.58
S9/124057	G11_1	100.00	1.95	6.34	2.31	4.28
S9/124057	G11_2	100.00	1.11	3.09	0.91	3.17
S9/124057	G11_3	100.00	0.81	7.19	0.40	2.49
S9/124057	G11_4	100.00	12.67	28.84	4.81	8.00
S9/125315	G21_1	100.00	0.83	4.39	0.64	4.58
S9/125315	G21_2	100.00	5.66	13.19	5.88	18.69
S9/125315	G21_3	100.00	5.53	12.10	1.77	4.42
S9/125315	G21_4	100.00	11.16	40.09	3.89	12.78

TABLE B.3
Official HO-Cap object identities for the evaluated groups; suffix order is _1 through _4.

Group	Official object descriptions in suffix order
G02	whole milk carton; cooked ham; chocolate drink powder; sauce bottle
G04	flapjack mix; chocolate fudge; herring fillets can; crackers box

Group	Official object descriptions in suffix order
G05	vegetable oil spread; body lotion; peanut chocolate box; sponge
G06	DVD; pumpkin creamer; yellow mustard; dish brush
G07	game controller; hot cocoa mix; dandruff shampoo; toy axe
G09	candy box; toy car; toothpaste box; chocolate syrup bottle
G10	soup mix; gel toothpaste; chocolate biscuit sticks; body wash
G11	coconut water; baby powder; baking soda; chocolate bar
G15	men's body wash; dandelion tea; cooking spray; joy controller
G16	toilet cleaner; laundry detergent; small coconut water; fabric softener sheets

Group	Official object descriptions in suffix order
G18	green tea latte mix; projector remote; right shoe; left shoe
G19	electric screwdriver; blue spatula; deodorant; ping pong paddle
G20	cappuccino mix; mustard bottle; toilet cleaner; dog toy bone
G21	moisturizing lotion; playing cards; pink spatula; blue brush
G22	chocolate bar; mayomust sauce; soup mix; gray spatula